



OECD Science, Technology and Industry Working Papers
2025/25

Assessing the relevance
of R&D funding towards
societal goals: Insights from
new data sources and AI-
assisted methods

Leonidas Aristodemou,
Silvia Appelt,
Brigitte van Beuzekom,
Fernando Galindo-Rueda

<https://dx.doi.org/10.1787/bafcdc7b-en>

OECD Science, Technology and Industry Working Papers

OECD Working Papers should not be reported as representing the official views of the OECD or of its member countries. The opinions expressed and arguments employed are those of the authors.

Working Papers describe preliminary results or research in progress by the author(s) and are published to stimulate discussion on a broad range of issues on which the OECD works. Comments on Working Papers are welcomed, and may be sent to Directorate for Science, Technology and Innovation, OECD, 2 rue André-Pascal, 75775 Paris Cedex 16, France.

Note to Delegations:

This document is also available on O.N.E Members & Partners under the reference code:

DSTI/STP/NESTI/MARIAD(2024)3/FINAL

© OECD 2025



Attribution 4.0 International (CC BY 4.0)

This work is made available under the Creative Commons Attribution 4.0 International licence. By using this work, you accept to be bound by the terms of this licence (<https://creativecommons.org/licenses/by/4.0/>).

Attribution – you must cite the work.

Translations – you must cite the original work, identify changes to the original and add the following text: *In the event of any discrepancy between the original work and the translation, only the text of the original work should be considered valid.*

Adaptations – you must cite the original work and add the following text: *This is an adaptation of an original work by the OECD. The opinions expressed and arguments employed in this adaptation should not be reported as representing the official views of the OECD or of its Member countries.*

Third-party material – the licence does not apply to third-party material in the work. If using such material, you are responsible for obtaining permission from the third party and for any claims of infringement.

You must not use the OECD logo, visual identity or cover image without express permission or suggest the OECD endorses your use of the work.

Any dispute arising under this licence shall be settled by arbitration in accordance with the Permanent Court of Arbitration (PCA) Arbitration Rules 2012. The seat of arbitration shall be Paris (France). The number of arbitrators shall be one.

Abstract

This paper presents and demonstrates a novel approach for assessing the relevance of public R&D funding towards societal goals. The approach relies on the development of a new machine learning classification model for R&D activity descriptions, trained on researchers' self-assessments about the relevance of their R&D activity towards Sustainable Development Goals (SDGs). Applied to R&D project descriptions in the OECD Fundstat database, the model shows how funding, including for basic research, contributes towards societal goals. The analysis allows to compare funding portfolios and provides evidence of the interdependencies between societal goals served by R&D funding and its potential to contribute to multiple goals.

Authors: Leonidas Aristodemou, Silvia Appelt, Brigitte van Beuzekom, Fernando Galindo-Rueda

Keywords: Research and Development (R&D), Government R&D funding, Machine Learning, Language Models, Sustainable Development Goals (SDGs), Survey data, Classification, Directionality

JEL codes: O38, O32, C38, C45, Q55

Acknowledgements

This report has been prepared by Leonidas Aristodemou, Silvia Appelt, Brigitte van Beuzekom and Fernando Galindo-Rueda at the Science and Technology Policy (STP) Division in the OECD Directorate for Science, Technology, and Innovation (DSTI).

The study has been conducted as part of the Programme of Work and Budget of the Committee for Scientific and Technological Policy (CSTP), under the auspices of the OECD Expert Group on the Management and Analysis of R&D and Innovation Administrative Data (MARIAD), established by the Working Party of National Experts on Science and Technology Indicators (NESTI) to coordinate and oversee work on R&D and innovation administrative data. Earlier versions of the document were presented for discussion in October 2024 at the NESTI and MARIAD official meetings, and in October 2025 at the CSTP official meeting.

The authors would like to express their gratitude towards a long list of contributors, primarily MARIAD, NESTI and CSTP delegates for their constructive feedback. In addition, some of the analysis in this study would not have been possible without the assistance of MARIAD delegates from several countries in making R&D project-level funding data available for analysis. The authors are particularly grateful for comments from CSTP, NESTI and MARIAD bureau members Miguel Gonçalves, Christian Naczinsky, Martine Garnier-Rizet, Yujin Jeong, Michael Lam, Roberta Pattono and Naohiro Shichijo, as well as MARIAD delegates Pascale Dengis, Marie-Alexandra Neouze and Kristof Van Crielingen. In addition, the authors would like to acknowledge several OECD colleagues. Daniel Sanchez Serra enabled use of the 2021 OECD International Survey of Science data under strict confidentiality and privacy requirements. Fabien Verger and Guillaume Kpodar provided useful comments on GBARD and the OECD RDS and OECD MSTI datasets. Petra Kelly provided valuable guidance on Energy and Planet-related R&D funding indicators. Stephanie Cairns and Kuniko Matsumoto contributed to the initial scoping of the SDG classification work. Colleagues from the OECD Statistics and Data Directorate (SDD) (Eric Anvar and Samuel Pinto Ribeiro), and the OECD Centre on Well-being, Inclusion, Sustainability and Equal Opportunity (WISE) (Liva Stokenberga and Ziga Zarnic), kindly reviewed the document and provided valuable additional feedback.

Selected results from this work are also featured in the [OECD Measuring Science and Innovation for Sustainable Growth report](#), with supplementary material available on [the OECD Fundstat webpage](#). Further information on the data and methods used in this analysis can be requested to mariad@oecd.org.

Table of contents

Abstract	3
Acknowledgements	4
Table of contents	5
Executive summary	7
1 Introduction	9
2 Mapping the contribution of science, technology and innovation to societal goals	11
2.1. Perspectives for assessing the directionality and relevance of STI towards societal goals	11
2.2. SDGs and related work on SDG classification for STI activity	12
2.2.1. Using SDGs to measure societal goals of R&D	12
2.2.2. General SDG mapping approaches for STI efforts	13
2.2.3. Keyword and AI assisted classification approaches	14
3 Data sources for assessing relevance	16
3.1. Classification training data	16
3.1.1. OECD International Survey of Science (ISSA)	16
3.1.2. Scopus bibliometric database on scientific publications	16
3.2. Classification evaluation data	17
3.2.1. OpenAlex database on scholarly metadata	17
3.2.2. OSDG community dataset on SDG-labelled text excerpts	18
3.3. Analysis data	18
3.3.1. OECD Fundstat database on R&D funding awards	18
3.3.2. Scopus bibliometric database on scientific publications	20
4 Developing a new classifier for the relevance of R&D to goals	21
4.1. Model development	21
4.2. Model evaluation	23
4.2.1. Alignment of self-assessed with other science SDG classifiers	23
4.2.2. Evaluation with generic SDGs tagged content	24
5 Results	25
5.1. Relevance of R&D funding and scientific publications towards goals	25
5.1.1. Results across funding sources	25
5.1.2. Mapping the R&D funding landscape according to societal goals	27
5.1.3. Profiling of funding sources by societal goals	28
5.1.4. Experimental results by geographical areas	31
5.1.5. Relevance trends for R&D funding and scientific publications	33
5.2. SDG R&D topics and interdependencies	38
5.2.1. Topic modelling of R&D projects relevant to Energy and Planet SDGs	38
5.2.2. Interdependencies in R&D funding between SDGs	40

6 Concluding remarks 41

References 43

Annex A. The OECD Fundstat database 50

A.1. Data sources in the OECD Fundstat database	50
A.2. Benchmark analysis of the OECD Fundstat database	52
A.3. Comparing the thematic orientation of R&D funding in Fundstat and GBARD	53

Annex B. Supplementary methodological information 55

Annex C. Supplementary results 58

Tables

Table 2.1. Mapping of socio-economic objectives (SEOs) for R&D to SDG thematic clusters	14
Table 2.2. SDG classification methods in the literature	15
Table 3.1. Most relevant SDG reported by 2021 ISSA respondents engaged in research activities	17
Table 3.2. R&D funding in the OECD Fundstat database, 2015-2023	19
Table 5.1. Estimates of R&D projects, R&D funding and scientific publications by SDG & SDG group	34
Table A.1. Data sources in the OECD Fundstat database, 2015-2023	50
Table B.1. Distribution of labels of sustainable development goals (SDGs) across the datasets	55

Figures

Figure 2.1. Conceptualising and measuring the direction of R&D funding towards societal goals	11
Figure 4.1. Comparison of SDG self-assessed classification with OpenAlex textual content classifier	24
Figure 5.1. SDG group distribution of R&D funding by source, 2015-2023	26
Figure 5.2. Two-dimensional structure of the R&D funding landscape according to SDG relevance	27
Figure 5.3. Thematic structure and proximity of R&D funding devoted to different SDGs, 2015-2023	29
Figure 5.4. Visualising and comparing R&D funding portfolios for selected UKRI bodies, 2015-2023	30
Figure 5.5. SDG Distribution of Fundstat R&D project funding by country/area, 2015-2023	32
Figure 5.6. SDG group distribution of R&D project funding by country/area, 2015-2023	33
Figure 5.7. SDG group distribution of scientific publications and funding awards' counts and values	35
Figure 5.8. Scientific publications, R&D funding and R&D projects by SDG group, 2015-2023	36
Figure 5.9. Trends in shares of R&D funding, awards and publications by SDG group, 2015-2023	37
Figure 5.10. Topics in R&D projects relevant to Energy and Planet related SDGs, 2020-2023	39
Figure 5.11. SDG relevance of R&D funding, by dominant SDG group in R&D awards, 2010-2023	40
Figure A.1. Coverage of government R&D funding in the 2024 OECD Fundstat database, 2010-2023	52
Figure A.2. Share of R&D funding by theme in Fundstat and GBARD for 19 OECD countries, 2021	54
Figure B.1. Out-of-sample SDG label classifier prediction on the OSDG dataset (n=8 768)	57
Figure C.1. SDG distribution of R&D projects, R&D funding and scientific publications, 2015-2023	58
Figure C.2. Distribution of R&D funding and R&D projects by SDG, 2015-2023	59
Figure C.3. Distribution of R&D funding and projects by SDG & data source, 2015-2023	60
Figure C.4. Distribution of R&D funding and projects by SDG & data source, 2015-2023 (continued)	61
Figure C.5. SDG group distribution of R&D projects by data source, 2015-2023	62
Figure C.6. Total R&D funding, R&D projects and scientific publications by SDG, 2015-2023	63

Boxes

Box 4.1. Language models for societal goals classification	22
--	----

Executive summary

What do billions of government investment in research and development (R&D) across OECD countries contribute to? Traditionally, such question was only addressed through the prism of established classifications at a high level of aggregation. Available statistics, whilst providing a comprehensive benchmark, leave unanswered several questions about the relevance of R&D funding towards specific and evolving societal goals.

To respond to such needs, the OECD has developed the Fundstat analytical infrastructure to develop and demonstrate approaches for working with administrative R&D microdata and AI tools for analysis of quantitative and qualitative data in conjunction with traditional methods. This paper presents a decisive step in the Fundstat initiative towards developing tools that can help provide evidence on the directionality and relevance of R&D funding.

Evidence on the potential contribution of R&D to different societal goals is crucial for assessing the balance and orientation of R&D and innovation policies. Yet, providing informative indicators remains challenging because support for R&D can influence the attainment of such goals through multiple and often indirect channels. AI-based classification methods applied to administrative data, while promising, are often reliant on superficial text features and tend to equate lexical similarity with relevance. Moreover, the widespread absence of explicit “goal” language in R&D activity descriptions leads to low measured rates of SDG relevance. Such approaches may understate the relevance of R&D, particularly basic and some forms of applied research, to outcomes that indirectly advance societal objectives.

To overcome these limitations, this study introduces an AI-based approach that is trained on effective domain-informed assessments of goal relevance for R&D. Close to 2 000 participants in the 2021 OECD International Survey of Science (ISSA2021) reported which Sustainable Development Goal (SDG), or none, they contributed to with their research. As it touches on virtually all areas of government policy, the UN SDG framework provides a convenient and relevant use case in the classification of unstructured R&D activity by goals. The survey data helped label a corpus of over 11 000 academic publications, which served in turn as the training dataset for fine-tuning SciBERT, a pre-trained domain-specific transformer-based language model (trained on a large corpus of scientific literature), into predicting goal relevance probability for a given text description. This yielded a model capturing relationships between research content and relevance towards the 17 SDG objectives, as well as the absence of SDG relevance.

The classifier, deployed on the Scopus custom database of scientific publications and the latest version of the OECD Fundstat, provides a basemap for the R&D landscape comprising approximately 2.1 million government-funded R&D project awards across 19 OECD countries and the European Union (EU) programmes implemented by the European Commission (EC), accounting for approximately USD 1.5 trillion in public R&D funding, between 2015-2023.

The results indicate that over half of total R&D funding between 2015-2023 is estimated to be relevant to two Sustainable Development Goals: SDG 9 (Industry, innovation and infrastructure: 30%) and SDG 3 (Good health and well-being: 23%). These goals dominate both in terms of funding volume and project counts. SDG 4 (Quality education: 8%), SDG 7 (Affordable and clean energy: 5%), and SDG 13 (Climate

action: 4%) follow as the next most prominent areas of R&D funding. Notably, only a small share of R&D funding (3%) is classified as having no SDG relevance (6% for scientific publication output), much less than estimated by previous literature. Whilst this might overstate genuine impacts on goals, these new results are more plausible since the UN SDG framework, SDG 9 (Industry, innovation and infrastructure) contains total R&D expenditure (as proportion of GDP) as an indicator towards target 9.5. As a result, any R&D activity could in principle contribute to SDG 9. This reflects a significant degree of ambiguity in the SDG framework with respect to R&D and innovation activity, which is not present in measurement frameworks designed for classification purposes.

Over one-third (37%) of total R&D funding recorded in Fundstat over the 2015–2023 period is found to be attributable to the “Prosperity” cluster of SDGs (SDGs 8, 9, 10, 11), followed by “Health” (SDGs 1, 2, 3, 5) at 29%, and “Planet” (SDGs 6, 12, 13, 14, 15) at 15%. Given their clear financial opportunity cost, budgeted or awarded amounts provide more meaningful measures of policy priorities than simple counts of activities or outputs, such as the number of funded projects or related scientific publications, also offering valuable insights into the breadth and intensity of research activity. Considering all three dimensions together helps to account for differences in cost structures across research domains, where some areas, such as health or infrastructure, are typically more capital-intensive than others. Viewed jointly, these measures provide a more comprehensive picture of the directionality of R&D funding and its alignment with societal goals. Recent trends also point to a shift towards fewer but larger R&D awards, as well as modest realignments in priorities in response to the COVID-19 crisis.

This paper also explores the advantages of granular data and AI tools to examine the R&D topics being funded within projects that score highly on environmental and energy goals. An unsupervised topic modelling exercise reveals research themes attracting public investment, with the topic structure is consistent with the selected goals (projects exhibit combinations of energy and environmental topics) and the types of project descriptions that underpin projects relevant to such goals (project descriptions exhibit different combinations of topics that describe the nature of the R&D, such as scientific topics and technology domains, on the one hand, and application areas that relate to goals, on the other).

The results also shed light on the synergies between different goals. Awards that primarily related to a given SDG also make very significant contributions towards other goals. For example, R&D projects primarily related to (economic) “Prosperity” goals (e.g., innovation and economic growth) often also contribute to environmental and health goals. Health-related R&D projects are in turn more narrowly focused but still show links to Planet and Prosperity related goals. Projects with environmental related goals in the SDG cluster “Planet” are particularly cross-cutting with R&D funding contributing to several other SDGs. These findings reflect the interconnected nature of societal challenges and highlight the potential for R&D investments to support multiple goals at once.

These results are no substitutes for studies of causal impact of R&D on outcomes; impact requires separate ex-post outcome analysis and counterfactuals. These new findings and methods are expected to equip policymakers, R&D funders, and other relevant stakeholders with evidence to support both strategic and operational purposes. In addition to extending the scope of Fundstat, possible avenues for future work include extending the application of this SDG classifier to other scientific data sources and exploring its use in mapping policy-relevant research portfolios, such as in health, climate, or energy domains, and assessing in more detail the orientations of support for emerging technologies.

1 Introduction

Science, technology, and innovation (STI) play a pivotal role in addressing global challenges by documenting their nature, scale and features, as well as by facilitating new solutions that overcome critical and hard to accept trade-offs under the state of the art of knowledge and technology. Evidence on the contribution of research and development (R&D), i.e., activity aiming to create and develop brand new knowledge and solutions (OECD, 2015^[1]), to a wide range of societal goals is crucial for assessing the balance and orientation of STI policies. It also plays a key part in communicating the return to public investments in R&D. Yet providing informative indicators and analysis on the goals of R&D is challenging because of the multiple channels through which support for R&D is intended to and ultimately impacts on the attainment of such goals.

OECD's long-standing work on classifying government budgets for R&D across societal goals focuses on assessing government's overarching aims as the result of the political process of setting the national budget. National experts report data to the OECD according to internationally agreed standards, matching budgetary lines to a defined range of "socioeconomic" objectives that closely match government functions like defence and health, and which include public goods such as the general advancement of knowledge (OECD, 2015^[1]). However, several policy questions require identifying what publicly funded R&D can be ultimately relevant for, regardless of the high-level stated aims and independently of any given classification framework. This requires having flexible allocation and classification mechanisms as well as data enabling granular analysis to serve different user needs. The OECD has recently developed the Fundstat analytical infrastructure with such an objective, working with R&D project award level microdata and using AI tools for analysing unstructured text descriptions of such awards. Examples of recent applications of Fundstat include studies that focus on both content, relevance and directionality of R&D funding, such as artificial intelligence (AI) development (Yamashita et al., 2021^[2]) and COVID-19 (Aristodemou et al., 2023^[3]).

In the absence of formal requirements for disclosing in a structured fashion what STI activities are relevant for, the key conceptual and practical measurement challenge is identifying goals from information available about STI activities, such as data on inputs from R&D project funding awards, or outputs such as scientific publications. This stands in contrast with patented inventions, where requests for protection must disclose their potential applications, claims which are further assessed by expert independent patent examiners. AI-based analysis of text descriptions needs to be adequately guided (i.e., trained) to reflect not only on the content of the STI activity but also consider how knowledge outputs can be key inputs for other STI activities that ultimately contribute to societal goals. For example, a summary description of research into properties of new materials may not necessarily disclose that such properties may play a critical role for applications in areas such as health, defence or energy. An appropriate degree of domain expertise is thus a necessary input into the training of reliable AI-based classification methods.

This paper uses the United Nations (UN) Sustainable Development Goals (SDGs) framework as a demonstration use case for a novel classification approach for R&D goals. The SDGs were adopted in 2015 to focus action towards addressing the world's most urgent economic, social, and environmental challenges. The 2015 OECD Daejeon Declaration on Science, Technology and Innovation (STI) Policies for the Global and Digital Age (OECD, 2015^[4]) underscored the importance of Science, technology, and innovation in achieving sustainable development, emphasising that STI is not just an engine of economic

growth but also a powerful tool for fostering sustainability and social well-being. Nearly a decade later, the 2024 OECD Declaration on Transformative Science, Technology, and Innovation Policies for a Sustainable and Inclusive Future (OECD, 2024^[5]) reinforced the necessity of transformative STI policies to address the SDGs, urging countries to reorient their STI policies to better address systemic issues like climate change, inequality, and resource depletion (OECD, 2024^[6]; OECD, 2024^[7]).

SDG allocation and classification methods used to date, typically based on keyword-matching or natural language processing (NLP), rely heavily on surface-level text features and equate lexical similarity with conceptual alignment. Recent studies have mapped the contribution of STI to the SDGs (Kashnitsky et al., 2024^[8]), by applying transformer-based SDG classifiers to scientific publications. As they have been applied to indicators of STI activity such as scientific publications' abstracts, the widespread lack of explicit "SDG" language in R&D activity descriptions tends to result in low rates of SDG relevance which understate the potential contribution of R&D, especially basic research and research applied to outcomes that may indirectly contribute to the goals.

This study introduces an AI-based probabilistic classification approach that is trained on effective domain-informed assessments of goal relevance for research. The applied model captures relationships between research content and relevance towards the 17 SDG objectives, as well as the absence of SDG relevance. The classifier has been deployed on the Scopus custom database of scientific publications (v.1.2024) and the latest version of the OECD Fundstat database (v.2024), which comprises approximately 2.1 million government-funded R&D project awards across 19 OECD countries and the European Union (EU) programmes implemented by the European Commission (EC), between 2015 and 2023. These awards accounted for approximately USD 1.5 trillion in government R&D funding over this period. This dual application to scientific publications and R&D awards data provides a view of R&D relevance towards societal goals, proxied using the SDG framework.

The remainder of the paper is organised as follows:

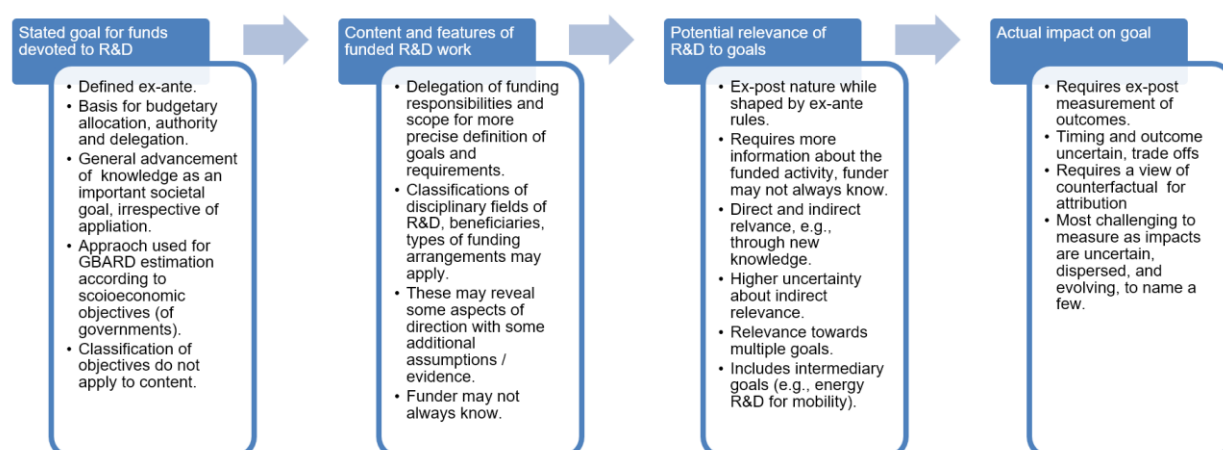
- Section 2 introduces the conceptual and empirical approaches adopted in mapping the contributions of science, technology and innovation (STI) to societal goals, taking the Sustainable Development Goals (SDGs) as reference framework. This is followed by a review of existing SDG classification methods, including keyword-based and AI-assisted approaches.
- Section 3 presents the data sources used for model development, evaluation, and analysis.
- Section 4 details the methodology for developing the SDG relevance classifier applied to R&D-related text. It discusses model training and evaluation, including validation with external SDG-labelled content and alignment with existing classification systems.
- Section 5 applies the SDG classifier to the Fundstat and Scopus databases to assess the relevance of R&D funding awards and scientific publication output. It also provides exploratory evidence on the thematic structures and interdependencies in R&D funding across different societal goals.
- Section 6 concludes with some potential implications and suggestions for future work.

2 Mapping the contribution of science, technology and innovation to societal goals

2.1. Perspectives for assessing the directionality and relevance of STI towards societal goals

There are different ways to conceptualise and measure the direction and relevance of R&D funding towards policy goals, contrasting the *ex-ante* or planned directionality (i.e. the intention to direct) and effective *ex-post* directionality (the attribute of having a direction) of R&D funding. Based on related OECD work documenting the multifaceted nature of directionality (OECD, 2023^[9]; OECD, 2023^[10]; OECD, 2024^[11]), Figure 2.1 outlines those by distinguishing between the measurement of stated policy goals, content and features of the funded R&D work, its potential relevance to the policy goal, and the ultimate impact of funded R&D activity on the policy goal.

Figure 2.1. Conceptualising and measuring the direction of R&D funding towards societal goals



Source: OECD own compilation

Present official measures of R&D directionality principally capture top-down budgetary processes that reflect ex-ante directionality. For classification purposes, they draw on the “Nomenclature for the analysis and comparison of scientific programmes and budgets” (NABS) and originally developed for STI measurement purposes but closely inspired by generic classifications of functions of government. The focus is on top-level policy societal or economic goals of R&D funding, irrespective of the *content* of the funded R&D and its potential relevance towards secondary goals, since the aim is allocation of financial

resources (budgeted amounts of potentially expenditure) towards mutually exclusive categories to avoid double counting in the reporting of Government Budgets for R&D (GBARD). Analysis of these data highlights that a major share of GBARD in many OECD countries falls under the category of General advancement of knowledge, which implies a low degree of ex-ante directionality towards specific societal goals. However, this does not imply that R&D budgets have a low degree of relevance towards such goals.

To analyse the content and characteristics of R&D support, other classifications are required, encompassing elements such as target beneficiaries (e.g., firms, universities), support conditions (e.g., type of STI activity), and type of instrument (e.g., grants, tax incentives) (OECD, 2015^[1]; OECD, 2023^[10]; OECD/Eurostat, 2018^[12]). While such information does not directly indicate stated goals, it can provide insights into implied direction. For example, R&D support targeted at start-ups in the pharmaceutical sector may suggest an underlying objective related to health or SME development. When combined with additional contextual information or assumptions, content-based classifications may support inferences about the ex-ante policy intentions behind R&D funding interventions. This information may not be known by the top-level budgetary authorities, who often grant a degree of decision-making autonomy in the use of funds not only to beneficiaries, such as universities, but also to intermediary bodies such as funding bodies, and those in turn to funding committees and panels.

Many policy questions require identifying what goals publicly funded R&D may be ultimately relevant for, regardless of the stated high-level objectives or the classification system applied. Assessing relevance to goals goes beyond stated intent and demands more detailed information about the nature and potential applications of the funded R&D, which government ministries and R&D and innovation funding bodies may not always have at their disposal. R&D activities can contribute to multiple societal goals, both directly and indirectly, including through intermediary applications or spillovers into other domains. Capturing this multidimensional relevance requires flexible classification approaches, access to granular data, and the integration of domain expertise, particularly when using AI-based methods for text analysis. Such models must be trained not only to interpret the immediate content of R&D activities but also to infer potential downstream contributions.

Assessing the impact of the funded R&D work on policy goals represents the final and most challenging element in measuring the direction of STI towards goals. It relies on the measurement of ex-post R&D, innovation and broader socio-economic outcomes and the ability to establish causal links between funded R&D work and those outcomes. The level and timing of the outcomes of this R&D work are highly uncertain ex-ante. Moreover, due to limited data availability and/or coverage, trade-offs may need to be made in the scale and scope of measurement and analytical efforts with a view to addressing the key policy questions at hand. To gauge the causal impact of R&D on the policy goal, the outcomes of interest would ideally be directly compared with their level in the counterfactual scenario in which no R&D support is provided. A fundamental estimation challenge arises as it is not possible to observe the “counterfactual”, i.e., how much R&D would R&D support recipients have performed had they not received this support. Being unable to observe the counterfactual directly, the best alternative is to compare R&D performers that receive support with those that do not but *are otherwise as similar as possible*, a strategy also adopted in related OECD distributed analysis on the impact of public support for business R&D (OECD, 2023^[13]; OECD, 2020^[14]).

2.2. SDGs and related work on SDG classification for STI activity

2.2.1. Using SDGs to measure societal goals of R&D

The sustainable development goals (SDGs) were adopted by the United Nations (UN) in 2015 as a comprehensive framework designed to address the world’s most urgent economic, social, and environmental challenges, setting a broader, more inclusive vision for sustainable development. The framework comprises 17 interrelated goals that cover areas ranging from poverty, education, and gender

equality to climate action and sustainable economic growth. Together, the SDGs reflect a universal call to end poverty, protect the planet, and ensure prosperity for all by 2030, emphasising the need for a balanced and collaborative approach to global development and economic growth. The global SDG indicator framework was developed by the Inter-agency and Expert Group on SDG Indicators (IAEG-SDGs) and adopted by the UN General Assembly in 2017 with a view to guiding the development of indicators and statistical data to monitor progress towards the SDGs and inform decision making in this area (UN, 2017^[15]). Each SDG is accompanied by targets that provide measurable steps toward achieving the broader goal (UN, 2024^[16]; UN, 2024^[17]). These targets are further monitored through 231 indicators, enabling countries and international organisations to track progress on each goal (UN, 2024^[18]). Building on the UN framework, the OECD has developed a methodology to measure the distance that OECD countries need to travel to achieve the SDG targets (OECD, 2019^[19]; OECD, 2025^[20]). A dedicated OECD hub on SDGs (OECD, 2025^[21]) provides access to the OECD SDG indicator infrastructure to inform countries about their position in achieving the SDGs and highlight critical data gaps that hinder their effective implementation. The SDG framework effectively covers a comprehensive range of societal goals that resembles existing classifications of generic functions of government. For this reason, and given its widespread recognisability, it provides a relevant test case in classification towards societal goals in general and for monitoring R&D efforts in particular.

2.2.2. General SDG mapping approaches for STI efforts

The positioning of Science, Technology, and Innovation (STI) within the SDG framework is a complex one. Since the headline indicator of gross domestic R&D expenditure is one among several used to monitor SDG 9 “Industry, innovation and infrastructure”, in principle all government (and private) funding of R&D then contributes to SDG 9. However, STI efforts can contribute towards any of the thematic goals rather beyond SDG 9, so a more interesting question is how government funding for R&D contributes to the different SDGs. For this reason, the OECD Directorate for Science, Technology and Innovation publishes within its GBARD statistics a set of indicators that cluster data by socioeconomic objectives into four clusters that are thematically related to groups of SDGs, tentatively mapped and named as Health and Society (OECD, 2025^[22]), Industry and Knowledge (OECD, 2025^[23]), Planet and Infrastructure (OECD, 2025^[24]), and Security (OECD, 2025^[25]) (Table 2.1).

Table 2.1 provides an overview of the tentative mapping of socio-economic objectives (SEOs) to four SDG related clusters, including a tentative list of thematically related SDGs for each cluster. GBARD measurement appears capable of approximating the SDG framework to some extent, in the case of objectives related to Health (NABS07; SDG 3), Education (NABS09; SDG 4), Energy (NABS05; SDG 7) and Environment (NABS02; SDGs 6, 12, 13, 14, 15) which can be directly connected to their SEO counterparts. However, several SEO-SDG correspondences are rather ambiguous. While various SDGs (SDG 6, 12, 13, 14, 15) can be incorporated under one SEO in the case of Environment (NABS02), the reverse applies in the case of SDG 9 (Industry, innovation and infrastructure) which appears to be thematically related to both industry (NABS11) and infrastructure (NABS04) related socio-economic objectives. Some SDGs such as SDG 1 (No poverty), SDG 2 (Zero hunger), SDG 5 (Gender equality), and SDG 10 (Reduced inequalities) also remain unmapped, being thematically more distant and not directly linkable to the SEO framework applied in reporting R&D budgets. The complexity of correspondences between classifications of policy objectives, combined with the fact that R&D funding may be provided with a view to achieving multiple, possibly interdependent policy goals, highlights the need of a flexible, AI assisted approach that draws on domain expertise in mapping the relevance of funded R&D work to policy goals such as the UN SDGs.

Table 2.1. Mapping of socio-economic objectives (SEOs) for R&D to SDG thematic clusters

SDG related clusters	Socio-Economic Objectives (SEOs) in the NABS classification
Health and Society • SDG 3 (Good health and well-being) • SDG 4 (Quality education) • SDG 16 (Peace, justice and strong institutions)	'Education', 'Culture, recreation, religion and mass media', 'Political and social systems, structures and processes', 'Health', 'General advancement of knowledge: R&D related to Medical and health sciences' (NABS 09-11, 07, 123 and 133)
Industry and Knowledge • SDG 8 (Decent work and economic growth) • SDG 9 (Industry, innovation and infrastructure)	'Industrial production and technology', 'General advancement of knowledge: R&D related to Natural sci., Eng. and technology, Social sci. and Humanities and the arts' (NABS 06, 121, 122, 125, 126, 131, 132, 135, 136); and R&D tax incentives
Planet and Infrastructure • SDG 6 (Clean water and sanitation) • SDG 7 (Affordable and clean energy) • SDG 11 (Sustainable cities and communities) • SDG 12 (Responsible consumption and production) • SDG 13 (Climate action) • SDG 14 (Life below water) • SDG 15 (Life on land)	'Exploration and exploitation of the Earth', 'Exploration and exploitation of space', 'Transport, telecommunication and other infrastructures', 'Environment', 'Energy', 'Agriculture', and 'General advancement of knowledge: R&D related to Agricultural and veterinary sciences' (NABS 01-05, 08, 124 and 134)
Security • SDG 16 (Peace, justice and strong institutions) • SDG 17 (Partnerships for the goals)	Defence (NABS 14)

Note: Mapping of the classification of socio-economic objectives (SEOs) based on NABS 2007 onto SDG groups. For some countries, field-based subcategories of 'General advancement of knowledge' (NABS12) are not reported, preventing potential alignment with SDG groups other than "Industry and knowledge", which includes under SDG 9 the target of increasing R&D investment.

Source: OECD (2025^[26]).

2.2.3. Keyword and AI assisted classification approaches

Several methodologies have been developed at the micro level to map the contribution of scientific research (Confraria, Ciarli and Noyons, 2024^[27]), technologies (Massucci and Seri, 2022^[28]; WIPO, 2024^[29]; Hajikhani and Suominen, 2022^[30]) and policy documents (Wincott, 2024^[31]) to the UN SDGs. Approaches have centred on keyword-based and artificial intelligence (AI) methodologies, each offering distinct advantages and limitations (Kashnitsky et al., 2024^[8]). Benchmarking studies comparing various SDG classification approaches, including those developed by Aurora and Elsevier, have shed light on the strengths and weaknesses of each method (Armitage, Lorenz and Mikki, 2020^[32]; Kashnitsky et al., 2024^[8]). While keyword-driven models generally excel in precision, they underperform in recall, i.e., they fail to capture the full spectrum of research contributions. In contrast, AI-based models offer better recall but introduce interpretability challenges. The combination of keyword-based and AI-driven methods is considered to represent the current state-of-the-art (see Table 2.2) (Kashnitsky et al., 2024^[8]).

Keyword-based methods, traditionally the most common and easiest to implement, utilise Boolean searches to identify relevant research by filtering specific sections of publications, such as titles and abstracts, for SDG-aligned keywords. Elsevier, Aurora, and a few others tailored SDG-specific queries that incorporate controlled vocabularies and expert-validated keyword combinations. Despite the utility of keyword-driven approaches, they are inherently limited by the complexity of SDG-related themes. Keywords can have multiple meanings depending on context. For instance, a publication on "clean energy" could be relevant to SDG 7 (Affordable and clean energy), SDG 13 (Climate action), or both, making it difficult to assign a single, definitive SDG label. Moreover, keyword approaches struggle to capture research that indirectly contributes to the SDGs, as they rely heavily on key term matching, often missing broader contributions (Purnell, 2022^[33]; Bautista-Puig et al., 2021^[34]; Romero Goyeneche et al., 2022^[35]; Boudt, Inghels and Spithoven, 2024^[36]).

In response to these limitations, AI methods, including machine learning (ML), transformer-based language models, and a variety of others, have emerged as promising complementary approaches to SDG

classification (Wulff, Meier and Mata, 2024^[37]; LaFleur, 2023^[38]; Ottaviani and Stahlschmidt, 2024^[39]). These methods, integrated into systems developed by organisations such as Elsevier and Aurora, have enhanced the ability to capture a broader range of SDG-related research outputs (Vanderfeesten, Jaworek and Keßler, 2022^[40]; Schmidt and Vanderfeesten, 2021^[41]). Aurora's SDG classifier, for instance, is utilised by OpenAlex, an open-source bibliographic database (Vanderfeesten, Jaworek and Keßler, 2022^[40]). Since 2018, Elsevier's SDG-specific search queries have been continuously refined using ML methods, improving recall while maintaining high precision (Bedard-Vallee Alexandre, James Chris and Roberge Guillaume, 2023^[42]; Roberge and James, 2025^[43]). These models are now employed in assessments such as the Times Higher Education Impact Rankings, which evaluate universities based on their research contributions toward the SDGs (Elsevier, 2022^[44]). Similarly, OSDG 2.0 combines ML models and ontology-based keywords to classify texts into the SDGs (Pukelis et al., 2022^[45]), while also OpenAI's GPT model was used alongside similarity measures to classify publications into SDGs (Yin et al., 2023^[46]). Additionally, Bayesian network models have been proposed as a method to explore the interrelationships between SDGs (Requejo-Castro, Giné-Garriga and Pérez-Foguet, 2020^[47]).

Table 2.2. SDG classification methods in the literature

Method	Selective references	Description	Results evaluation	Advantages	Limitations
Keyword/ontology-based methods (A)	(Armitage, Lorenz and Mikki, 2020 ^[32] ; Purnell, 2022 ^[33] ; Schmidt and Vanderfeesten, 2021 ^[41])	Traditional method using Boolean searches to filter titles and abstracts with SDG-aligned keywords. Queries developed by Elsevier, Aurora, and others.	• Scientific publication dataset with expert labels	• High precision	• Low recall • Struggles with interdisciplinary research • Context-based keyword variations
AI-related methods (B)	(Wulff, Meier and Mata, 2024 ^[37] ; LaFleur, 2023 ^[38] ; Pukelis et al., 2022 ^[45] ; Ottaviani and Stahlschmidt, 2024 ^[39] ; Vanderfeesten, Jaworek and Keßler, 2022 ^[40] ; Yin et al., 2023 ^[46])	Utilises machine learning (ML) and language models in supervised and unsupervised settings to classify research under relevant SDGs.	• Scientific publication dataset with expert labels	• Broad coverage • Multiple languages • Enhanced recall • Used in global rankings • Analysis of large datasets	• Complexity • High computational resources
Combination of (A) + (B)	(Pukelis et al., 2022 ^[45] ; Bedard-Vallee Alexandre, James Chris and Roberge Guillaume, 2023 ^[42])	Combine both AI-related methods and ontology-based keyword matching	• Crowd sourcing • Scientific publication dataset with expert labels	• Supports multiple languages • High precision	• Still under development • Refinement for multilingual adaptation

Note: The primary method is reported in the table and often references deploy more than one method within each category.

Source: OECD own compilation, based on (Kashnitsky et al., 2024^[8]).

Whilst SDG classification could potentially inform resource allocation (Roy et al., 2018^[48]; IPCC, 2021^[49]; IPCC, 2022^[50]; IEA, 2021^[51]; Borchardt et al., 2023^[52]), it is not straightforward to implement for STI activity. Classification efforts tend to identify research content with goals. This distinction is crucial, as current classifiers focus heavily on the textual content, what the research is about, but doing so, risk to miss out on the utility or societal relevance of the work. Upstream basic research is difficult to assess based on content alone, as its goal relevance might not be immediately obvious. Several recent classifiers rely on text-based methods that classify research based on abstracts or textual descriptions of the work, with expert bibliometricians' validating the results but not necessarily the research subject-experts themselves. While these systems provide valuable insights, they tend to systematically be on the side of caution, significantly understating potential relevance. The discrepancies between self-assessed relevance and automatic text-based classification underscore the need for clearer formulations of relevance measurement objectives and the effective mechanism to bring expert input into the analysis.

3 Data sources for assessing relevance

3.1. Classification training data

3.1.1. OECD International Survey of Science (ISSA)

The primary training dataset to build the SDG classifier is based on responses to the 2021 OECD International Survey of Science (ISSA2021) (Bello and Galindo-Rueda, 2020^[53]; OECD, 2024^[54]; OECD, forthcoming^[55]). The survey collected data from over 3 000 scientific researchers, who were asked a range of questions on working conditions, societal engagement and research relevance, career prospects, and other topical subjects like the impact of the COVID-19 pandemic (OECD, 2024^[54]). Under the societal engagement and research relevance theme, given widespread awareness of the SDG framework, respondents were asked about the most relevant SDG for their research activity. The question put forward was the following: “*You may be aware of the United Nations Sustainable Development Goals (SDGs) set in 2015 by the United Nations. In your opinion, does your work relate, directly or indirectly, to any of the following goals? Please select the most relevant SDG*”. Respondents were able to select only one of the 17 SDGs or the option of “Not relevant towards any SDG”. Close to 95% of participants (2 911 individuals) responded to the question. While ISSA2021 was not designed to be representative of a given research-connected population, as it used non probabilistic sampling methods, its value for the present study stems from its ability to provide an SDG labelled database for research purposes through record linking. Accordingly, ISSA2021 is primarily used as a labelling instrument (via record-linkage to publications) rather than a vehicle for population inference.

3.1.2. Scopus bibliometric database on scientific publications

From the 2 911 individuals that responded to the research SDG relevance question, approximately 75% of the respondents (2 171 individuals) were matched to their author profiles in the Scopus custom database (v. January 2024) through ORCID IDs, enabling the obtention of information about the research conducted by respondents. Table 3.1 outlines the number of respondents to the 2021 ISSA survey, which responded to the SDG relevance question and were matched as publication authors to Scopus by each SDG.

The matched distribution highlights that 26% of the respondents selected SDG 3 (Good health and well-being) as the most relevant SDG for their research, followed by 14% for SDG 9 (Industry, innovation, and infrastructure), and approximately 13% for SDG 4 (Quality education). Analysis of the responses by authors indicating SDG 4 (Quality education) as objective reveal that these do not necessarily correspond to research on education but to scholarship-driven research. This goal is more commonly reported by researchers in Higher Education, who have teaching duties, involved in basic research and especially in the fields of Arts and Humanities and Social Sciences. Approximately 8% of respondents indicated that the research is not directly relevant to the 17 SDGs. This is then followed by SDG 7 (Affordable and clean energy) with 6%, and SDG 12 (Responsible consumption and production), SDG 15 (Life on land) and SDG 13 (Climate action) with 5% each. Socially oriented SDGs, such as SDG 5 (Gender equality), SDG 17 (No

poverty reduction), and SDG 1 (No poverty), were less frequently selected, each accounting for about 1% of responses. The record linking exercise does not appear to have a material impact on the distribution of SDGs, with some relative differences (e.g., record linking increases the relative role of SDG 3 by 3 points, while it reduces that of SDG 4 Education by 2 points). Whilst the ISSA survey was not designed for scientific researchers alone, this appears to be the dominant group given the high matching rate. A broader ranging survey that had captured a higher percentage of researchers in business would probably have resulted in a higher SDG 9 share, and record linking to scientific publications then would have reduced that percentage in the labelled database.

Table 3.1. Most relevant SDG reported by 2021 ISSA respondents engaged in research activities

SDG no.	SDG label	ISSA2021 respondents	Share (%)	ISSA2021 respondents matched to Scopus	Share (%)
3	Good health and well-being	675	23.2	566	26.1
9	Industry, innovation, and infrastructure	419	14.4	307	14.1
4	Quality education	431	14.8	276	12.7
7	Affordable and clean energy	146	5.0	130	6.0
12	Responsible consumption and production	136	4.7	115	5.3
15	Life on land	116	4.0	105	4.8
13	Climate action	124	4.2	98	4.5
11	Sustainable cities and communities	109	3.7	76	3.5
14	Life below water	84	2.9	73	3.4
6	Clean water and sanitation	59	2.0	54	2.5
2	Zero hunger	60	2.1	51	2.3
10	Reduced inequalities	67	2.3	41	1.9
8	Decent work and economic growth	72	2.5	36	1.7
16	Peace, justice, and strong institutions	69	2.4	30	1.4
5	Gender equality	57	1.9	26	1.2
17	Partnership for development	27	0.9	16	0.7
1	No poverty	20	0.7	7	0.3
0	Not relevant to any SDG	240	8.2	164	7.6
Total		2 911	100.0	2 171	100.0

Note: The table shows the SDG relevance distribution (profiles across the 18 SDG relevance options - 17 SDGs and none) among ISSA2021 respondents to the SDG item (94% of all respondents), and among those, those who were successfully matched as publication authors to their respective Scopus Author (75% of the total response).

Source: OECD analysis of the 2021 OECD International Survey of Science (ISSA).

Over the period 2000–21, over 11 000 publications authored or co-authored by these respondents were retrieved from Scopus using their matched Scopus author profile. Each publication was labelled according to the most relevant SDG (or alternatively, no SDG relevance) as reported by the linked author in the survey, providing the foundation for the training dataset.

3.2. Classification evaluation data

3.2.1. OpenAlex database on scholarly metadata

OpenAlex is an open scholarly metadata database that aggregates metadata on a wide variety of research outputs, including articles, books, datasets, and dissertations, while capturing relationships among authors, institutions, and citations. OpenAlex provides automated SDG classifications of scientific publications, generated through a machine learning transformer-based model developed by the Aurora Universities Network (Vanderfeesten, Jaworek and Keßler, 2022^[40]). The OpenAlex SDG Classifier is based on a

multilingual BERT (mBERT) model, which estimates the relevance of research outputs to each of the 17 SDGs using textual content such as titles and abstracts. Research outputs are tagged with SDG labels when the model assigns a prediction score above a defined threshold (0.40). The OpenAlex SDG labels are compared with ISSA2021 researchers' self-assessed SDG relevance. The comparison of machine-assigned labels to self-reported SDG classifications provides insights into the correspondence of automated text-based approaches in capturing the intended societal orientation of scientific research.

3.2.2. OSDG community dataset on SDG-labelled text excerpts

The OSDG community dataset is a public dataset that contains SDG-labelled text excerpts, coming from various sources, such as reports, policy documents, and scientific publication abstracts, with contributions from over 1 400 volunteers (OSDG; UNDP IICPSD SDG AI Lab; PPMI, 2024^[56]). The OSDG dataset does not contain only text from research-focused content, but other related content (policy, news etc.), which makes the out-of-sample test also an out-of-corpus test, i.e., evaluating the SDG classification model's ability to generalise across different types of text. Each text in the OSDG dataset is evaluated based on its relevance to one SDG and validated by at least 3 volunteers (Pukelis et al., 2022^[45]). To ensure that the trained SDG classification model is evaluated using high-confidence data, the OSDG dataset is used as an out-of-sample evaluation test, using only those texts: (i) with perfect agreement among volunteers (i.e., 100% agreement); (ii) without negative labels (rejections); and (iii) that 5 or more volunteers have agreed with the SDG label.

3.3. Analysis data

3.3.1. OECD Fundstat database on R&D funding awards

Fundstat is a novel data infrastructure focused on R&D funding awards developed by the OECD for analytical purposes. Administrative data on government R&D project funding offer potential for international comparative analysis (OECD, 2015^[1]). Award-level data, containing both quantitative (funding) and qualitative information (project/activity descriptions) are collected, maintained and often published by funding bodies for accountability and transparency. Their international compilation and analysis can provide insights into the ex-post directionality of government R&D funds allocated at project or equivalent level, with varying levels of robustness that depend on factors such as the extent to which award/project-based funding is representative of public funding R&D within a country as well as the degree of availability and completeness of available records. Fundstat has been used for the analysis of R&D funding for AI and to address the COVID-19 pandemic (Aristodemou et al., 2023^[3]; Yamashita et al., 2021^[2]).

The version of Fundstat (v. 2024) used for this paper comprises data on government R&D funding awards for R&D and innovation funding bodies in 19 OECD countries (Australia, Austria, Belgium, Canada, Czechia, Estonia, Finland, France, Germany, Japan, Ireland, Latvia, Lithuania, Norway, Portugal, Sweden, Switzerland, the United Kingdom, and the United States) and the European Union (EU) – European Commission (EC) programmes from 2015 onwards, which have made data available for the analysis through either open and restricted channels. Coverage varies widely across countries and sectors, depending on factors such as:

- The relative share of funding allocated via competitive projects vs. institutional block grants.
- The availability and openness of administrative R&D and innovation funding data from national and regional agencies and organisations.
- The extent of project data/metadata completeness (e.g., funding amount, project description).

Table 3.2. R&D funding in the OECD Fundstat database, 2015-2023

Country/ area	R&D project awards	(Of which) with missing funding information	R&D Funding (USD PPP million)	Award size (USD PPP million per project)	GBARD Coverage 2021 (%)	Selected factors contributing to reduced coverage (e.g., missing funding bodies, programmes, funding types, etc.)
AUS	37 731	169	23 530	0.63	72	• Mission R&D programmes
AUT	9 968	1 175	6 580	0.75	46	• Austrian Research Promotion Agency (FFG) • Climate and Energy Fund • Institutional funding (e.g., OEAW, ISTA, LBS)
BEL	58 802	16 603	8 169	0.19	22	• Institutional funding
CAN	354 864	3 679	55 776	0.16	95	-
CHE	37 625	1 060	15 712	0.43	49	• Institutional funding
CZE	16 287	1 253	15 836	1.05	61	• Czech Science Foundation (CSF) • Institutional funds (Czech Academy of Science)
DEU	80 418	476	85 059	1.06	37	• German Research Foundation (DFG) • Agency for Renewable Resources (FNR) • Federal Office for Agriculture and Food (BLE) • Federal Ministry of Defence (BMVg) • Institutional funding (e.g., MPS, FhG, HGF)
EST	5 385	240	1 562	0.30	72	• Institutional funding
EU-EC	159 096	12 723	208 327	1.42	-	• No EC GBARD available for comparison
FIN	5 203	33	2 462	0.48	28	• Government research institutes • Institutional funding
FRA	25 674	1 535	26 870	1.11	20	• Institutional funding • EPST (e.g., CNRS, INSERM, INRIA) • EPIC (e.g., CEA, ONERA, BRGM)
GBR	104 328	39 411	85 850	1.32	96	-
IRL	8 986	5 486	2 727	0.78	40	• Enterprise Ireland • IDA Ireland • Teagasc • Institutional funding
JPN	294 889	8 902	91 287	0.32	5	• Agencies (e.g., JST, NARO, JAXA) • Institutional funding (e.g., RIKEN, NIMS, AIST) • R&D programmes (e.g., Moonshot, SIP, WPI)
LTU	864	3	295	0.34	39	• Institutional funding
LVA	10 325	1 887	3 575	0.42	100	-
NOR	16 845	702	9 306	0.58	58	• Institutional funding • Innovation Norway • Enova
PRT	27 498	1 014	4 675	0.18	83	-
SWE	36 738	1 610	13 856	0.39	71	• Institutional funding • Swedish Environmental Protection Agency • Sector specific agencies
USA	856 150	26 480	834 806	1.01	70	• Part of Defence R&D not reported as projects • Govt. intramural R&D not covered by project funds
Total	2 147 676	124 441	1 496 262	0.74	51	

Note: Ordered in alphabetical order of column "Country/area". Percentage coverage (%) of R&D funding in the OECD Fundstat database by country quantifies the proportion of R&D funding accessible within that country's R&D data sources (Table A.1) in the form of R&D project awards, relative to the country's Government Budget Allocations for R&D (GBARD), excluding the General University Funds (GUF) (OECD, 2025^[57]; OECD, 2025^[58]). The reference year 2021 represents the most recent period for which both Fundstat and GBARD offer high completeness and granularity. The calculation of award size (in USD PPP million per project) excludes projects with missing or nil funding information. For Belgium (BEL), data on total R&D projects and funding is provided by the Belgian Science Policy Office (BELSPO), and by the Department of Economy, Science, and Innovation (EWI) of the Flemish government.

Source: OECD analysis of the Fundstat database (v.2024) and the Main Science and Technology Indicators (MSTI) database (OECD, 2025^[57]).

Table 3.2 reports on the within country coverage of the Fundstat database, providing information on key factors impacting on the representativity of the available data. Coverage is calculated as the share of R&D funding captured in Fundstat relative to each country's Government Budget Allocations for R&D (GBARD), excluding General University Funds (GUF). This benchmark focuses the comparison on the theoretical portion of public R&D funding that could be at best traced through project-level funding records. In 2021, the most recent year for which GBARD data offer high completeness and granularity and Fundstat data show stability in terms of completeness and no truncation, Fundstat captures approximately 51% of the targeted government R&D funding across the 19 participating countries. Country-level coverage varies widely, from high project traceability in the United Kingdom (96%) and Canada (95%), to low coverage in Japan (5%), France (20%), and Germany (37%). Countries with high coverage tend to strongly rely on competitive project-based funding. The use of centralised and open data reporting systems also contributes to observed differences. In contrast, countries with dominant institutional R&D models where large public research performing organisations such as CNRS or Fraunhofer receive direct block funding are relatively poorly covered in Fundstat. In addition to structural funding patterns, gaps in data provision also reduce coverage, several R&D funding and innovation bodies or ministries missing from the data sources in Fundstat (Table A.1). Incomplete data at the project level, another factor influencing reduced data coverage, is relatively limited: less than 1% of project awards (19 385 out of over 2.1 million) lack textual information in either the title or description, while about 7% of project awards (124 441 out of over 2.1 million) have no available funding information. These entries are typically linked to internal amendments, disclosure controls, or administrative placeholders and are excluded from average award size calculations.

To interpret Fundstat outputs appropriately, thematic benchmarking is conducted, comparing the distribution of funding across a selection of societal goals in Fundstat and GBARD, identifying potential alignment or divergence in R&D funding priorities (see Annex A). Fundstat represents a growing effort to increase transparency and analytical depth in the measurement of public R&D funding. While limitations in coverage persist, it complements traditional indicators by providing insight into the granular composition, directionality, and relevance of public R&D funding.

3.3.2. Scopus bibliometric database on scientific publications

In addition to its role in training the SDG classification model, the Scopus bibliometric database also serves as a core analysis dataset for the large-scale application of the classifier. Once the SDG classifier was trained using labelled data from the OECD 2021 ISSA survey, it was used to predict the SDG relevance of all scientific publications indexed in Scopus between 2000-2022. Each publication record includes structured bibliographic textual information such as titles, abstracts, which serve as input to the classifier that generates probabilistic predictions of relevance to each of the 17 SDGs and the No relevance label.

4

Developing a new classifier for the relevance of R&D to goals

This section describes the methodology and evaluation techniques adopted in the development of a novel classifier for assessing the relevance of R&D to societal goals, taking the UN SDGs as a reference framework. This new machine learning classifier, drawing on self-assessments of research relevance by scientific authors in the OECD International Survey of Science (ISSA2021) (Section 3.1.1) and textual content in scientific publications from the Scopus database (Section 3.1.2), predicts the SDG classification of R&D funding awards in the OECD Fundstat database and scientific publications.

4.1. Model development

Inferring information about R&D goals from scientific technical documents involves the analysis and classification of specialised scientific language that contains highly technical terminology, discipline-specific jargon, and complex sentence structures that differ from general language (Dan Clark et al., 2023^[59]). Furthermore, conceptual differences between R&D content and goals potentially served by the R&D activities also need disentangling. The SDG classification developed in this study relies on transformer-based language models, which underpin many of today's large language models (LLMs) and constitute an integral part of natural language processing (NLP), a branch of artificial intelligence (AI) that focuses on the interaction between computers and human languages (Jurafsky and Martin, 2023^[60]). NLP enables machines to process, analyse, and generate human language in a manner that is meaningful (Box 4.1).

Model development involved two main steps: training and validation. Training a model requires “teaching” it to recognise patterns in labelled data so it can make predictions on unseen data. Once the model was trained, its performance and generalisability were assessed based on an unseen (validation) data set. The dataset for training and validation came from the 2021 OECD ISSA survey (Section 3.1.1), where the scientific publications of the matched ISSA respondents to the Scopus author profiles had been labelled according to the 17 SDGs and a “No SDG relevance” label. The dataset contained 11 719 scholarly articles (OECD, forthcoming^[55]). To account of possible changes in relevance of keywords and textual content in scientific domains over time and the concentration of the ISSA respondents' scientific publication activity in the years 2016-20, the dataset was limited to the 2010-2021 period. The dataset was then split into the training set and the validation set consisting of 1 000 observations (Annex B). This split is essential to monitor overfitting and to evaluate how well the model generalises to unseen data. The text dataset was then tokenised using SciBERT's tokeniser, and each article's SDG labels were one-hot encoded.

The SciBERT model for multi-label classification (Beltagy, Lo and Cohan, 2019^[61]) was fine-tuned using transfer learning to predict the SDG for each article based on its content (Church, Chen and Ma, 2021^[62]; Wang and Chen, 2022^[63]; Pan and Yang, 2010^[64]). While the underlying training data structure reveals a single most relevant SDG per respondent, rather than responses to 17 different SDG questions, multi-label classification (i.e., SDG by SDG prediction) was applied following other related literature (Vanderfeesten, Jaworek and Keßler, 2022^[40]; Kashnitsky et al., 2024^[8]). In multi-label classification, the model assesses

the likelihood of each SDG being relevant to the project without considering the presence or absence of other SDGs, assuming that each label's probability is independent of the probabilities for other labels. To reconcile the different predicted probabilities for each SDG, a post-prediction ad hoc normalisation was applied. The raw probabilities of the 18 labels (17 SDGs and the “No SDG relevance” option in the survey) were re-scaled to sum to one, producing a normalised probability distribution across these 18 labels. This makes it possible to interpret these probabilities as relative weights of the project's contribution to each SDG and use them to apply R&D funding to different SDGs on a fractional basis to ensure additivity.

Box 4.1. Language models for societal goals classification

Transformer-based language models are pre-trained on large corpora of text data, and during this process, they learn general language patterns (Vaswani et al., 2017^[65]; Lecun et al., 1998^[66]; Devlin et al., 2019^[67]). They can then be fine-tuned on specific tasks or domains, which leads to improved performance in areas like scientific text classification. A well-known example is the Bidirectional Encoder Representations from Transformers (BERT) and its specialised variants like SciBERT, which have enabled machines to understand context, and handle domain-specific language. While BERT is pre-trained on general text data such as Wikipedia, its variant SciBERT is trained on a corpus of 1.14 million scientific papers from Semantic scholar. This makes SciBERT effective for tasks that involve academic articles, because of its specialised vocabulary focusing on scientific terminology and structures (Beltagy, Lo and Cohan, 2019^[61]). SciBERT's training on scientific literature enables it to better handle the technical vocabulary found in academic research (Ambalavanan and Devarakonda, 2020^[68]). For example, when analysing a scientific publication on advanced materials for renewable energy, a general language model might misinterpret the content due to its unfamiliarity with terms like “photovoltaic efficiency” or “perovskite solar cells”. On the other hand, SciBERT is better suited to grasp these scientific terms and correctly classify the article under SDG 7 (Affordable and clean energy).

Using SciBERT has many advantages:

- Domain-specific training: Unlike general language models, SciBERT has been trained on scientific texts, making it more adept at understanding specialised terminology.
- Contextual understanding: The transformer architecture allows SciBERT to understand not just the meanings of words, but also their relationships across sentences and paragraphs.
- Efficient fine-tuning: SciBERT is already pre-trained on general scientific texts, so fine-tuning it for SDG classification is faster and computationally efficient than training a model from scratch.

Source: OECD own compilation.

A challenge in this classification task is the imbalance of SDG labels. Some SDGs, such as SDG 3 (Good health and well-being), are more represented than others, such as SDG 5 (Gender equality). Training a model on an imbalanced dataset could lead to poor performance on the under-represented SDGs (Johnson and Khoshgoftaar, 2019^[69]). To counter this, two strategies were employed:

- Class weights: Class weights were computed based on the frequency of each SDG. By inversely scaling the weights according to the label frequencies, the model focused more on under-represented classes during training (Cui et al., 2019^[70]; Hu et al., 2025^[71]). The use of class weights ensured the model did not become biased towards the frequent classes and paid equal attention to under-represented SDGs.
- Focal loss: The focal loss gave higher importance to misclassified examples, helping the model to focus on hard-to-classify instances, by modifying the standard cross-entropy by down-weighting well-classified examples and focusing more on difficult ones (Lin et al., 2017^[72])

The model was trained through an iterative optimisation process where its internal parameters (network weights) are updated in response to the prediction error computed by the loss function. Each complete pass through the training dataset is known as an “epoch” (LeCun, Bengio and Hinton, 2015^[73]). Throughout training, the model aimed to minimise the discrepancy between its predictions and the true targets. Model performance and generalisability were assessed on a separate validation dataset (not seen during training) using standard metrics: accuracy, precision, recall, F1 score, and ROC AUC (Aristodemou, 2020^[74]).

At the end of training, the training loss, which quantifies the model's prediction error on the training data, was 0.003, while the validation loss, which quantifies the model's prediction error on the validation data, was 0.058, suggesting a slight overfit with the moderate difference showing that the model generalises acceptably. The resulting precision score was 0.73, meaning that 73% of the SDG predictions classified as relevant by the model were indeed correct. The recall score was 0.57, meaning that approximately 57% of true relevant SDGs were captured. The F1-score, which balances precision and recall, stands at 0.64 (64%), suggesting a fair trade-off between identifying relevant SDGs and avoiding false positives. While the model was effective at classifying common SDGs (as reflected by the high precision), the recall and the F1-score showed that there is still room for improvement to consistently capture all relevant SDG categories, especially less frequent SDGs. The ROC-AUC score of 0.78 (78%) indicated that the model systematically ranked correct predictions higher than incorrect ones, implying that it could differentiate between SDGs with reasonable confidence. Accuracy stood at 60%, and while this may appear as a modest performance, it is important to consider that there are 18 distinct labels, many of which are often thematically similar and intertwined.

4.2. Model evaluation

4.2.1. Alignment of self-assessed with other science SDG classifiers

Figure 4.1 compares the alignment between SDGs labels assigned by self-assessment of research relevance through the 2021 OECD ISSA survey and those predicted by the text-based classifier, as implemented in OpenAlex (Vanderfeesten, Jaworek and Keßler, 2022^[40]). This comparison covers 9 812 (out of the 11 719) scientific publications authored by respondents to the ISSA survey that were successfully matched to OpenAlex via digital object identifiers (DOIs). The row-wise normalised confusion matrix facilitates an overview of the degree of alignment and discrepancies (Priem, Piwowar and Orr, 2022^[75]; OECD, forthcoming^[55]). A confusion matrix evaluates the performance of a classification model by comparing the predicted labels to the true labels. Each row of the matrix corresponds to the actual classes, while each column represents the predicted classes. The diagonal elements indicate correct predictions, while the off-diagonal elements show misclassifications, revealing which classes are most frequently confused by the model. Each row represents a self-assessed SDG label, with the corresponding normalised values between 0 (no alignment) and 1 (highest alignment), indicating how publications within that category were labelled by the text-based system. Similarities are observed for certain SDGs: for example, SDGs 7, 9 and 14 demonstrate moderate agreement, indicating that both self-assessment and automated classification can converge in their identification of research relevance for some goals. At the same time, the matrix reveals that the two methods can yield different perspectives for other SDGs, especially when research is highly interdisciplinary or where relevance may be embedded in less explicit textual cues. For SDGs such as 1, 5, 8, 9, 12, and 13, a greater divergence in classification outcomes can be observed, suggesting that each method may capture distinct aspects of research relevance.

The presence of “no label” (NL) assignments in the text-based classifier implemented in OpenAlex provides a valuable perspective on SDG classification. In the OpenAlex approach, a publication is assigned NL when its prediction score is less than a threshold (e.g., 0.40). This is the case for 34% of publications, i.e., 3 348 out of 9 812 in the matched sample, which are marked with no label, and thus no SDG relevance. By contrast, self-assessment labels identified a lower percentage of “no label” publications, namely 8% of

publications, i.e., 780 out of 9 812 in the matched sample. This difference highlights the advantage of research relevance self-assessment: researchers are often able to recognise SDG relevance that may not be explicitly apparent in textual content alone, especially for interdisciplinary research topics.

Figure 4.1. Comparison of SDG self-assessed classification with OpenAlex textual content classifier

		SDG label based on textual content from OpenAlex																	
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	NL
SDG label based on responses from 2021 ISSA survey	1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.3
	2	0.0	0.2	0.1	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.3
	3	0.0	0.1	0.3	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4
	4	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4
	5	0.0	0.0	0.3	0.0	0.1	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.3
	6	0.0	0.0	0.0	0.0	0.0	0.3	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.3
	7	0.0	0.0	0.0	0.0	0.0	0.1	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5
	8	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.2	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.3
	9	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4
	10	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.2	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3
	11	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.3	0.0	0.1	0.1	0.1	0.0	0.0	0.2
	12	0.0	0.1	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.1	0.1	0.0	0.0	0.4
	13	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.5
	14	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.8	0.1	0.0	0.0	0.1
	15	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.4	0.0	0.0	0.2
	16	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.1	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.3	0.1	0.2
	17	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.3
	0	0.0	0.0	0.1	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5

Note: Comparison of SDG classification labels assigned to a matched sample of 9 812 scientific publications. Each publication was authored by a respondent to the 2021 OECD ISSA survey and successfully matched to an OpenAlex record using the publication's digital object identifier (DOI). For each publication, two SDG labels are compared: (i) self-assessment-based SDG label, derived from the survey respondent's own indication of the most relevant SDG for their research (as reported in the ISSA survey) (OECD, forthcoming^[55]); (ii) text-based SDG label from the automatic text SDG classification in OpenAlex (Priem, Piwowar and Orr, 2022^[75]) generated automatically by the OpenAlex classifier, which uses machine learning and textual analysis of the publication's content. Each row corresponds to an SDG as identified via self-assessment, and each column to the SDG assigned by the OpenAlex text classifier (including "NL" for cases where no SDG label was assigned). Values are normalised within each row and range from 0 (no alignment) to 1 (highest alignment). Individual respondents may be associated with multiple publications, so the same self-assessed SDG can be propagated to several publications from the same author from the ISSA survey. Higher values along the matrix diagonal reflect alignment between the two approaches, while off diagonal values indicate differences.

Source: OECD analysis of the 2021 OECD International Survey of Science (ISSA) and OpenAlex.

4.2.2. Evaluation with generic SDGs tagged content

Model evaluation is the process of assessing how well a trained model performs on new, "unseen" data. The aim is to determine whether the model generalises well beyond the training data, ensuring it can be reliably used. In this case, following training and validation, an out-of-sample test is performed on the OSDG community dataset. This dataset is used to evaluate the model and assess the effectiveness and accuracy of the SDG classifier in classifying text based on their alignment with the SDGs.

The key issue is whether the SDG classifier reflects the actual utility of the work for SDG goals, not just its content. Existing classifiers rely on content-based tagging, which often fails to capture real-world research relevance. The SDG classifier developed as part of this study is different, leveraging self-reported relevance data from the ISSA2021 survey to provide a more accurate assessment of how work contributes to SDG outcomes. It achieves a micro-F1 score of 0.54 and a macro-F1 score of 0.47 against the OSDG dataset (Kashnitsky et al., 2024^[8]), highlighting the added value of the relevance-based tagging (Annex B, Figure B.1).

5 Results

The section presents the main results from the analysis of SDG relevance of R&D funding at different levels of aggregation (funding source, country and overall results). While R&D funding awards descriptions differ in many ways from descriptions of scientific research outputs in scientific publications, several similarities exist, including their textual language, which supports the decision to apply the same classifier to funding awards (Aristodemou et al., 2023^[3]). In applying the SDG classifier to Fundstat and the Scopus databases, R&D projects and scientific publications are fractionally assigned to multiple SDGs based on the normalised probabilities. To minimise the extent to which changes in data coverage influence results, the SDG analysis focuses on the 2015-2023 period, where the coverage of Fundstat is stable.

When reviewing these results, it is important to keep in mind that cross-country differences and limitations in the data coverage of Fundstat relative to GBARD (Table 3.2) may shape the identified thematic patterns in R&D funding. To contextualise the analysis of R&D project awards by societal goals within the OECD Fundstat database, a benchmark comparison with GBARD by Socio-Economic Objective (SEO) (OECD, 2025^[57]) is available in the Annex material to this report.

5.1. Relevance of R&D funding and scientific publications towards goals

5.1.1. Results across funding sources

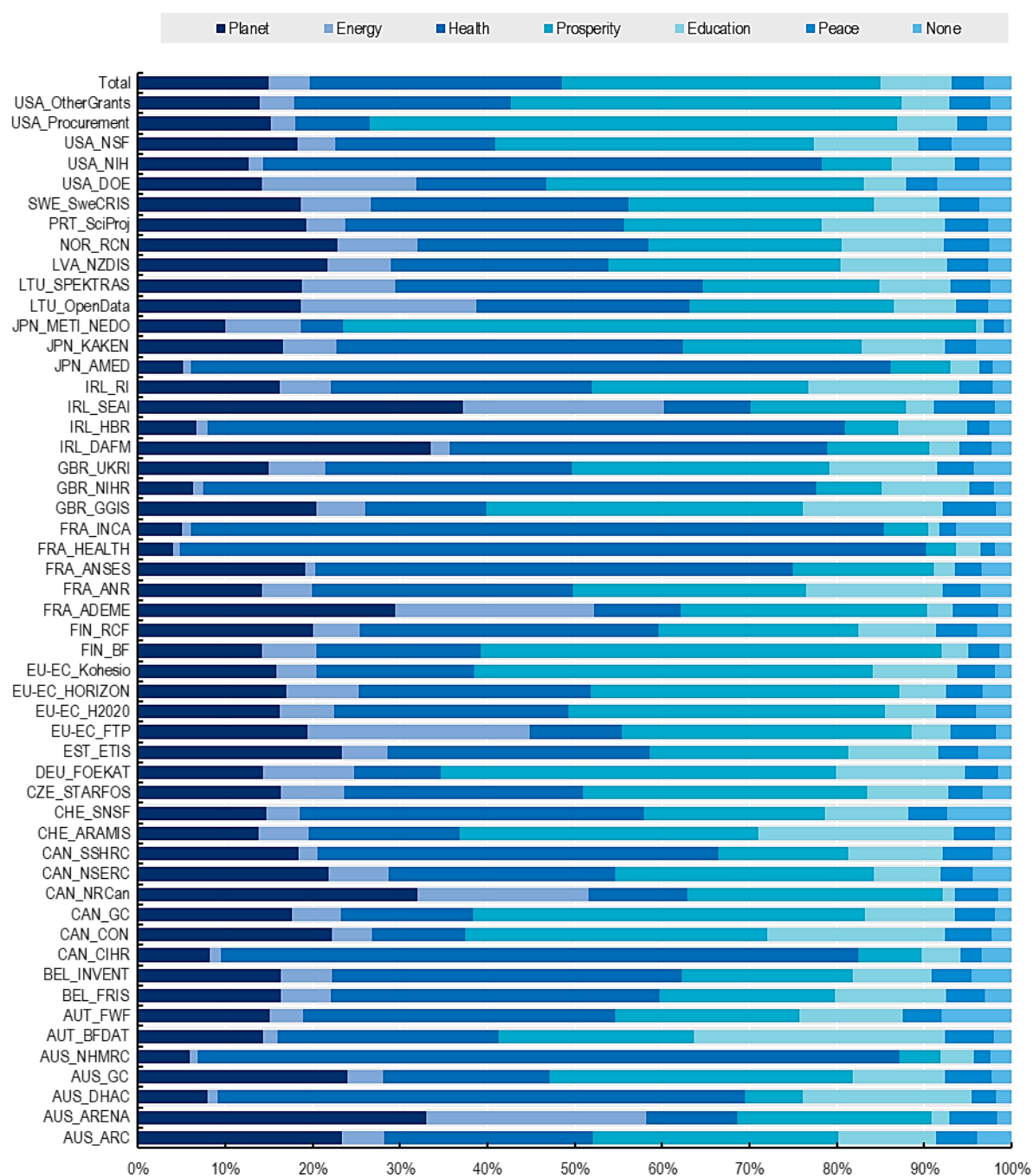
This section presents results on the relevance of R&D funding at the level of the national data sources (often corresponding to funding bodies) covered by the Fundstat database (see Table A.1 for the full descriptions of the abbreviations used in figures and text). Additional results on the distribution of R&D projects and R&D funding by SDG are presented in Annex C.

For each data source covered by Fundstat, Figure 5.1 displays the estimated distribution of R&D funding by SDG group. SDG groups Prosperity and Health account for the bulk of R&D funding across most national data sources, followed by SDG groups Planet and Energy. The SDG group Prosperity, predominantly characterised by SDG 9 (Industry, innovation, and infrastructure), consistently attracts large R&D funding from innovation-oriented data sources, e.g., CAN_GC (45%), FIN_BF (53%), JPN_METI_NEDO (72%), USA_Procurement (60%), EU-EC_Kohesio (46%) etc. The Health group, primarily captured by SDG 3 (Good health and well-being) dominates the R&D portfolios of health-focused funding bodies, with FRA_HEALTH (85%), AUS_NHMRC (80%), JPN_AMED (80%), FRA_INCA (79%), IRL_HBR (73%), CAN_CIHR (73%), GBR_NIHR (73%), and USA_NIH (64%). Funding towards SDG 7 (Affordable and clean energy) is particularly salient for USA_DOE (18%), AUS_ARENA (25%), IRL_SEAI (23%), and FRA_ADEME (23%). R&D funding directed to the SDG group Planet, comprising SDG 13 (Climate action), SDG 14 (Life below water), SDG 15 (Life on land), and SDG 12 (Responsible consumption and production) is most noticeable for IRL_SEAI (37%), followed by IRL_DAFM (34%), AUS_ARENA (33%), CAN_NRCAN (32%), and FRA_ADEME (29%). Other data sources with a strong environmental focus include NOR_RCN (23%), CAN_NSERC (22%), and EST_ETIS (23%).

SDG Education (SDG 4) is prominent in data sources with social science funding streams, e.g., AUS_BFDAT (29%), CHE_ARAMIS (22%), and IRL_RI (19%). The Peace group generally attracts lower

levels of R&D funding, ranging from 0% to 6% across data sources. A residual but important category is None, which captures the fractional share of each funding item's probability mass that is not allocated to any of the 17 SDGs. Given the post-prediction normalisation step for each project's 18 predicted probabilities to sum to unity, the None category reflects the residual relevance share not attributed to an SDG. Across the data sources, the None category accounts for approximately 1 %–8 % of R&D funding.

Figure 5.1. SDG group distribution of R&D funding by source, 2015-2023



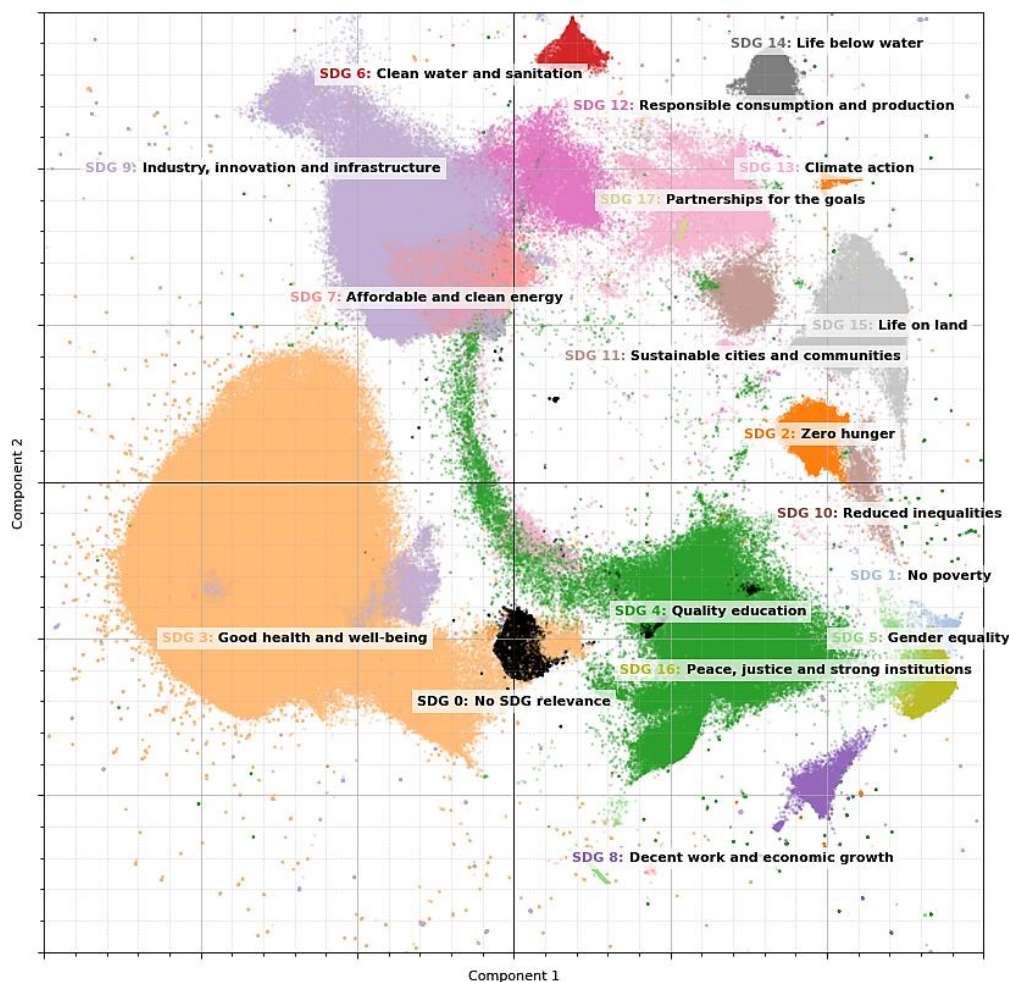
Note: See Table A.1 for more complete details on the funding sources covered in Fundstat corresponding the abbreviations in this figure. The analysis excludes projects with missing or nil funding information.

Source: OECD analysis of Fundstat database (v.2024).

5.1.2. Mapping the R&D funding landscape according to societal goals

Mapping tools help investigate the structure of R&D funding devoted to different SDGs. A dimensionality reduction method, Principal Component Analysis (PCA), has been applied to the estimated SDG probability profiles for each R&D project in the OECD Fundstat database. The mapping analysis projects the 18-dimensional SDG probability vector into data that can be visualised in 2 dimensions. Since each award is portrayed by vector with an estimated probability distribution over SDG relevance options, the probability values are first transformed through a centred log-ratio transformation. This normalisation enables the PCA to capture meaningful axes of variation in the probability structure. The PCA-reduced representation is then plotted in a two-dimensional map where projects with similar SDG probability profiles tend to orient and cluster together, while more distinct clusters are separated further away from the origin. Figure 5.2 depicts the resulting two-dimensional representation of the Fundstat R&D funding landscape, where each point on display corresponds to an award, coloured according to its dominant estimated SDG.

Figure 5.2. Two-dimensional structure of the R&D funding landscape according to SDG relevance



Note: The two-dimensional map visualises the SDG orientation of government R&D awards with funding information in the OECD Fundstat database. A compositional centre log ratio transform is applied to the principal component analysis performed on the probabilities predicted by the SDG classifier on government-funded R&D projects in the OECD Fundstat database. The PCA identifies two main dimensions as drivers of overall data variance, depicted as the main axes, and clusters SDGs according to their commonalities and distinctiveness. The two principal components explain 35.5% of the variance in the original data. The coloured points represent projects assigned to their dominant SDG label. Clustering reflects thematic relatedness of awards. The visualisation is based on a random sample of 1 million R&D awards.

Source: OECD analysis of the Fundstat database (v.2024).

The first two principal components of the principal component analysis (PCA) on display explain approximately 35.5% of the variance in the SDG relevance profiles for all awards. Principal components are not always straightforward to interpret as measures of specific quantities, but they help organise the funding landscape around key co-occurrence patterns. Several distinct clusters emerge when considering the spatial distribution of awards according to the dominant SDG. For instance, awards scoring highest on SDG 3 (Good health and well-being) form a large, cohesive block to the left/ lower left of the figure. Awards scoring high in SDG4 education and in “No relevant SDG” probabilities locate in the central region of the map, reflecting a lower degree of goal distinctiveness vis a vis the other goals. Indeed, education and training themes in the award descriptions that result in SDG4 classification co-occur with awards related to all other SDGs.

The top right quadrant hosts several SDGs focused on diverse aspects of environmental sustainability, namely SDG 12 (Responsible consumption and production), SDG 13 (Climate action), SDG 14 (Life below water), and SDG 15 (Life on land). The top left quadrant comprises SDG 9 (Industry, innovation and infrastructure) and SDG 7 (Affordable and clean energy). The latter appears to serve as connection between SDG 9 and the cluster of environmental sustainability SDGs on the top right quadrant, consistent with the pursuit of “double dividend” projects that aim to achieve both energy transition and industrial innovation objectives.

The bottom right of the chart displays the focal points of clustering for quality and inclusion-themed SDGs, namely SDG 1 (No poverty), the bulk of SDG 4 (Quality education), SDG 5 (Gender equality), SDG 8 (Decent work and economic growth), SDG 10 (Reduced inequalities), and SDG 16 (Peace, justice and strong institutions). The location of the Zero hunger SDG in between the environmental sustainability SDGs and SDG 10 reveals close connections between both themes

Like a world map, the limitations of flat 2D visualisation could be potentially overcome in higher dimensional spaces. For example, a spherical representation allowing connections between the most remotely positioned items would allow to show connections between bottom down located goals with those in the upper left, thus thematically connecting awards relevant to SDG8 on work and economic growth with those on SDG9 Industry. Likewise, awards relating to health would connect with environmental sustainability SDGs. These do not emerge in the 2D visualisations because the links are not as strong as those captured in the two main principal components.

5.1.3. Profiling of funding sources by societal goals

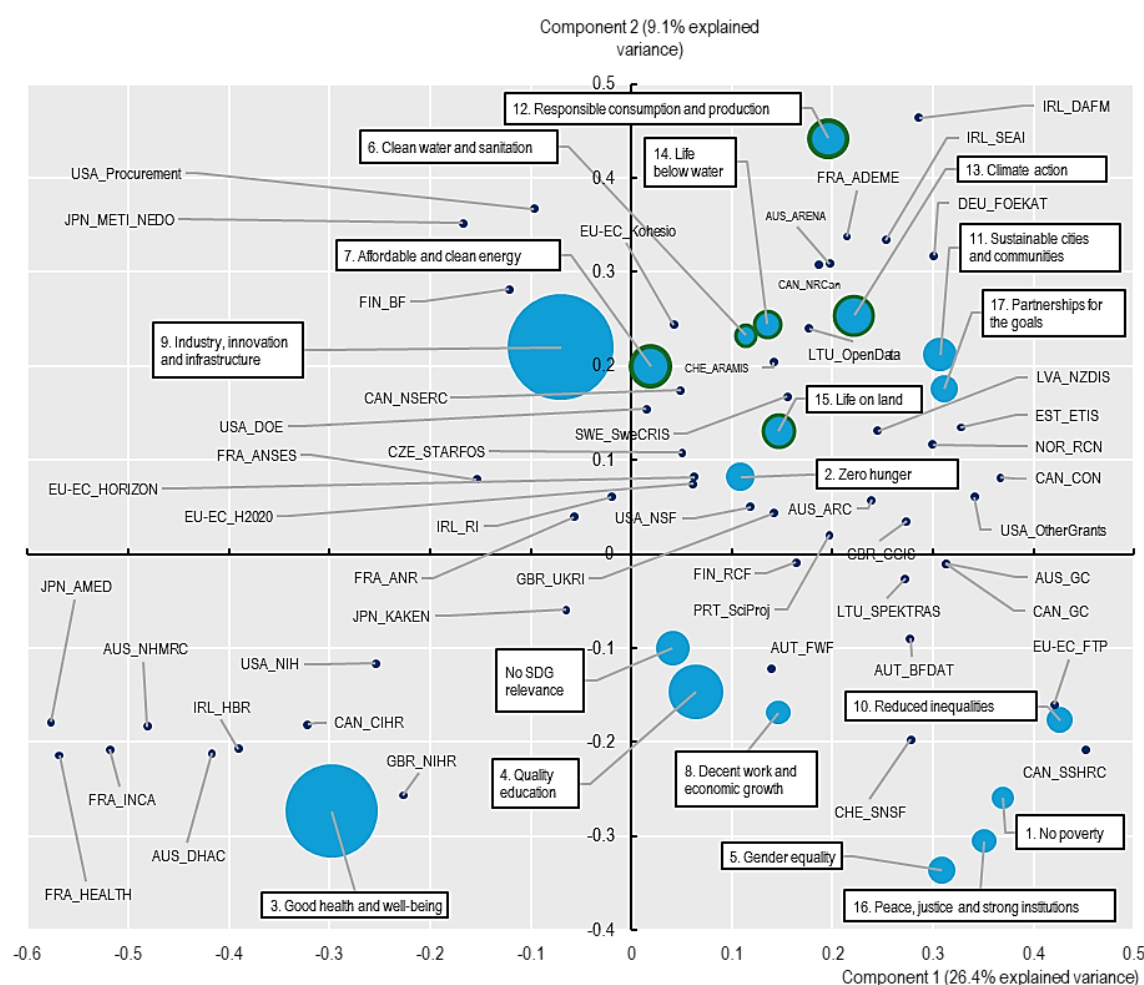
Using the methods and representation tools used for Figure 5.2, Figure 5.3 displays the central points for the different SDGs and the funding sources covered in Fundstat. Each light blue bubble in the biplot corresponds to SDG loadings on these two principal components and the “bubble” size is proportional to its share in the total amount of R&D funding in Fundstat for 2015-2023. The dark blue dots represent the funding bodies projected onto the very same space, whereby the coordinates of each R&D funding agency represent an R&D project award weighted average of the coordinates derived for the R&D projects managed by the agency in this two-dimensional space. In other words, the positions of SDGs and funding sources are overlayed on the SDG-relevance basemap for Fundstat awards.

The results highlight patterns of thematic R&D specialisation and orientation of funding bodies. For example, the environmental sustainability space in the top right quadrant is populated by sources corresponding to specialised bodies such as France’s Environment and Energy Management Agency and Ireland’s Department of Agriculture and Sustainable Energy Authority, but there are also sources relating to broader goals such as Germany’s Federal Ministry of Education and Research and Lithuania’s open R&D projects database. Japan’s METI-NEDO, Business Finland and USA Procurement are salient examples of funding sources that populate the top left quadrant that hosts SDG9. Conversely, the SDG 3 Health space is inhabited by health and biomedical R&D funding agencies such as the US National Institutes of Health, Japan’s AMED, the UK National Institute of Health Research and France’s Ministry of

Health. There appear to be relatively fewer funding sources located in the “Inequalities” cluster, such as Canada’s Social Sciences and Humanities Research Council and the Swiss National Science Foundation. Data sources encompassing funding for a broad range of goals cluster in the centre of the figure, such as the EU Horizon programme. The US National Science Foundation, Japan’s KAKEN and UK Research and Innovation are also found in this space with slight differences in orientation away from health topics.

Figure 5.3. Thematic structure and proximity of R&D funding devoted to different SDGs, 2015-2023

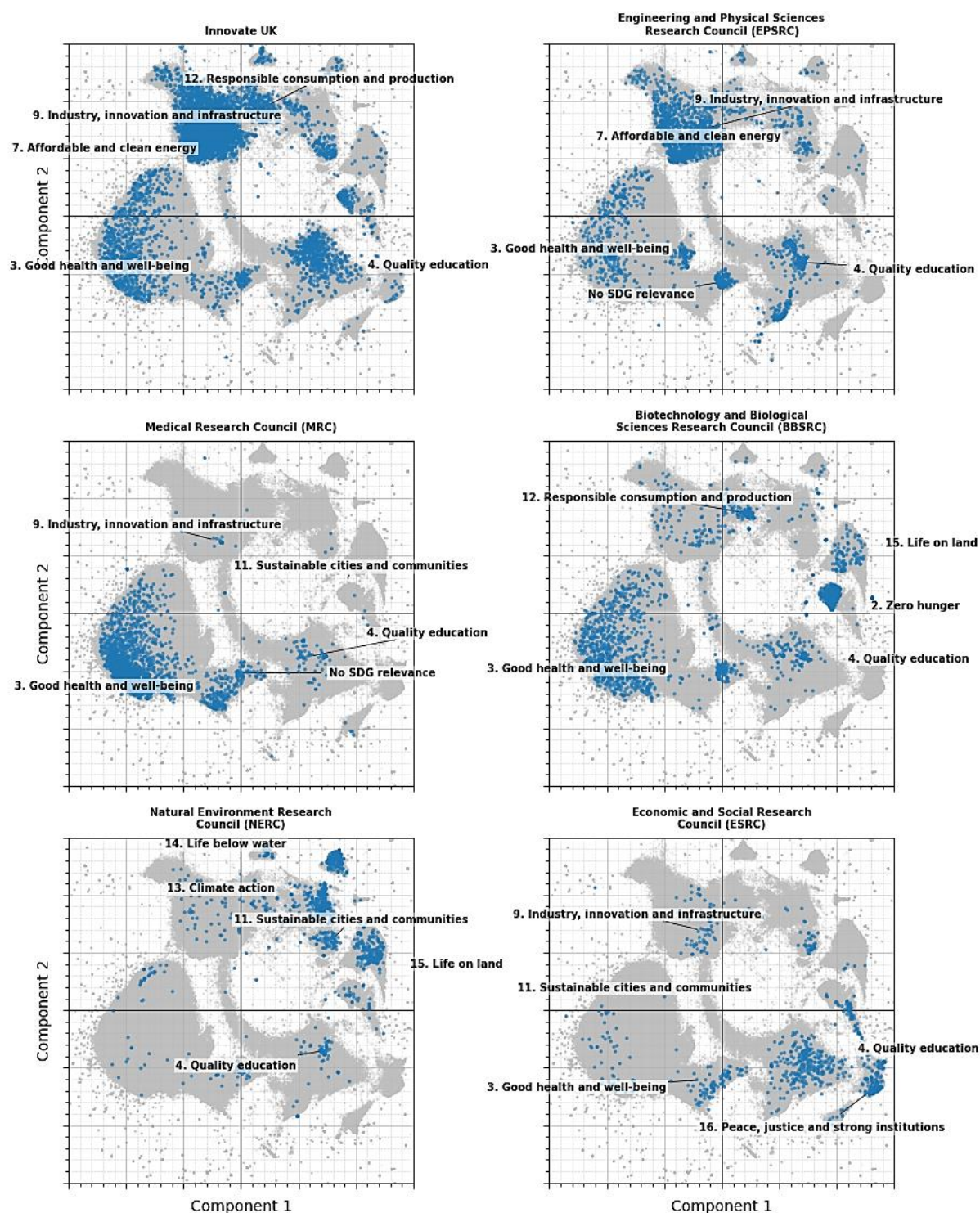
Principal component analysis (PCA) biplot of estimated SDG relevance probabilities



Note: The PCA is performed on the probabilities predicted by the SDG classifier on government-funded R&D projects in the OECD Fundstat database. The PCA identifies two main dimensions as drivers of overall data variance, depicted as the main axes, and clusters SDGs according to their commonalities and distinctiveness. Light blue shows the SDG loadings, and dark blue shows the data sources projected on the principal components space. Bubble size for the SDGs reflects the estimated share of total R&D funding for each SDG. The two principal components shown explain approximately 35.5% of the variance in the original dataset. Analysis limited to R&D projects with available funding information. Source: OECD analysis of the Fundstat database (v.2024).

Using the same base map of awards for all Fundstat, Figure 5.4 presents for illustrative purposes the distribution in the SDG space of R&D project awards for selected bodies within the United Kingdom Research and Innovation (UKRI), which was presented in the previous figure as a single point. The analysis conveys the informational value of disaggregated analysis enabled by a micro-based map and metadata about the funding awards in the UKRI database.

Figure 5.4. Visualising and comparing R&D funding portfolios for selected UKRI bodies, 2015-2023



Note: The two-dimensional map visualises the orientation of government-funded R&D projects in the OECD Fundstat database with respect to the 17 Sustainable Development Goals (SDGs) and the category “no SDG relevance” for a selected sample of UKRI’s research councils, using the basemap in Figure 5.3. The light grey background shows the full population of R&D projects, while the blue dots highlight the projects funded by UKRI’s specific councils shown in each panel. Labels indicate the major SDGs where projects are most concentrated.

Source: OECD analysis of Fundstat database (v.2024).

Awards provided by the six selected bodies are highlighted in dark blue dots over the overall landscape in grey dots. This visualisation illustrates how the research activities of individual UKRI bodies are distributed across the broader landscape of SDG-related research clusters. Prominent clusters are annotated with the SDGs most strongly represented in each council's portfolio. For example, Innovate UK and the Engineering and Physical Sciences Research Council (EPSRC) R&D projects are primarily concentrated in the space occupied by SDG 9 (Industry, innovation and infrastructure), SDG 7 (Affordable and clean energy), but are also present more sparsely in other areas. For instance, Innovate UK appears to have a significant presence (in terms of numbers of awards) in the Education space, through several awards that focus on developing technology-based learning tools, as well as in the health innovation space.

Funding by the Biotechnology and Biological Sciences Research Council (BBSRC) and the Natural Environment Research Council (NERC) are principally located in the environmental sustainability space but differ in their specific SDG orientation and in terms of connection with health. NERC scores high on SDG 13 (Climate action), SDG 14 (Life below water), and SDG 15 (Life on land), whilst BBSRC on SDG 2 (Zero hunger) and SDG 15 (Life on land) and SDG 3 (Health). The latter is indicative of multiple application areas for biological research. Funding provided by the Medical Research Council (MRC) is clearly positioned in the SDG3 space, with links to SDG4 Education. Awards provided by the Economic and Social Research Council (ESRC) feature across highlighted societal dimensions, with relatively high concentrations in SDG 16 (Peace, justice and strong institutions) and SDG 11 (Sustainable cities and communities) in addition to SDG4 Education.

Visualisation tools such as those provide in this example highlight how the goal relevance classification analysis can be used for providing intuitive comparisons of relevance profiles that, while these may often confirm funding agency mandates and calls, can also illustrate complementarities, as later discussed in section 5.2.

5.1.4. Experimental results by geographical areas

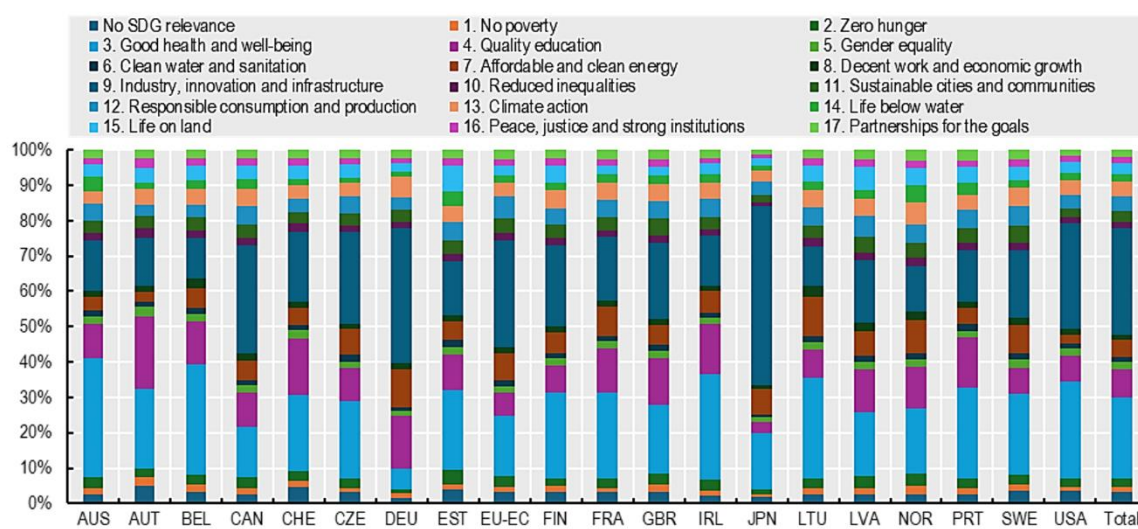
While differences in data coverage by country/area in the OECD Fundstat database (Table 3.2) call for caution when reviewing and comparing aggregated results, these are still worth examining as they convey information about policy priorities in the use of discretionary project-based R&D funding

Figure 5.5 shows that SDG 9 (Industry, innovation and infrastructure) and SDG 3 (Good health and well-being) account for a significant share of total R&D funding and R&D project counts over the 2015-2023 period across most of the economies on display. SDG 9 accounts for around 50% of R&D funding in Japan, and 30% in the United States and EC programmes. SDG 3 accounts for nearly 30% in the United States, and close to 15% in the case of Japan and EU-EC programmes. SDG 7 (Affordable and clean energy) accounts for the largest share in the R&D funding portfolio of Lithuania (11%), Germany (10%) and Norway (9%). Similarly to data at the level of funding source, estimates for project counts and funding amounts differ for several countries, owing to significant differences in the average funding amount per award.

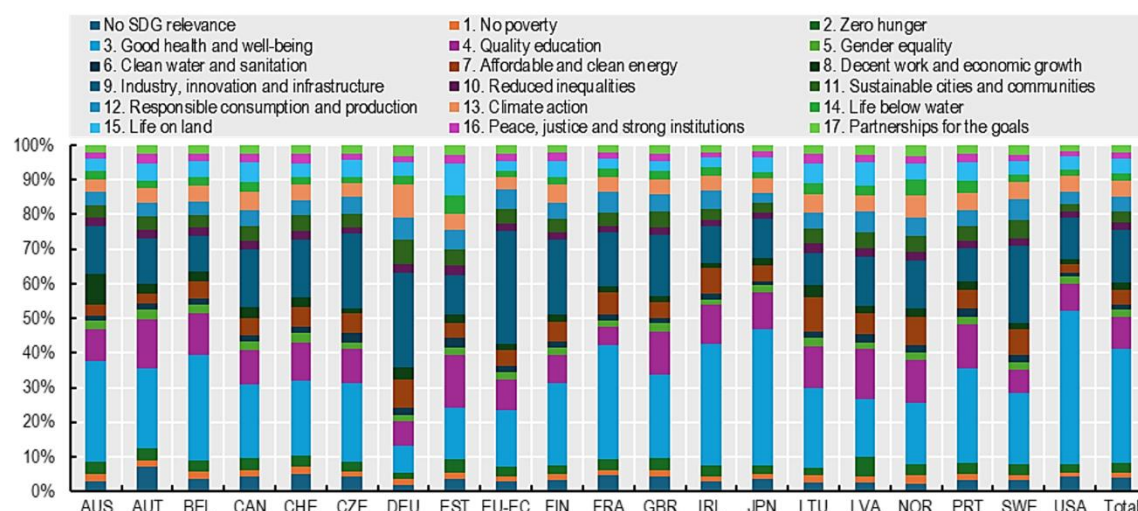
The results at SDG category level (Figure 5.6) show that the SDG groups of Prosperity and Health account for the largest share of R&D funding across countries (Figure 8a). Over 2015-2023, Prosperity represents a core priority area in the -available for analysis- R&D funding portfolios of Germany, United States, Japan, and EC. The share of R&D funding attributable to Prosperity ranges from 20% in Lithuania to 50% in Japan, while for Health this value ranges from 10% in Germany to 41% in Australia.

Figure 5.5. SDG Distribution of Fundstat R&D project funding by country/area, 2015-2023

(a) SDG distribution of R&D funding



(b) SDG distribution of R&D projects

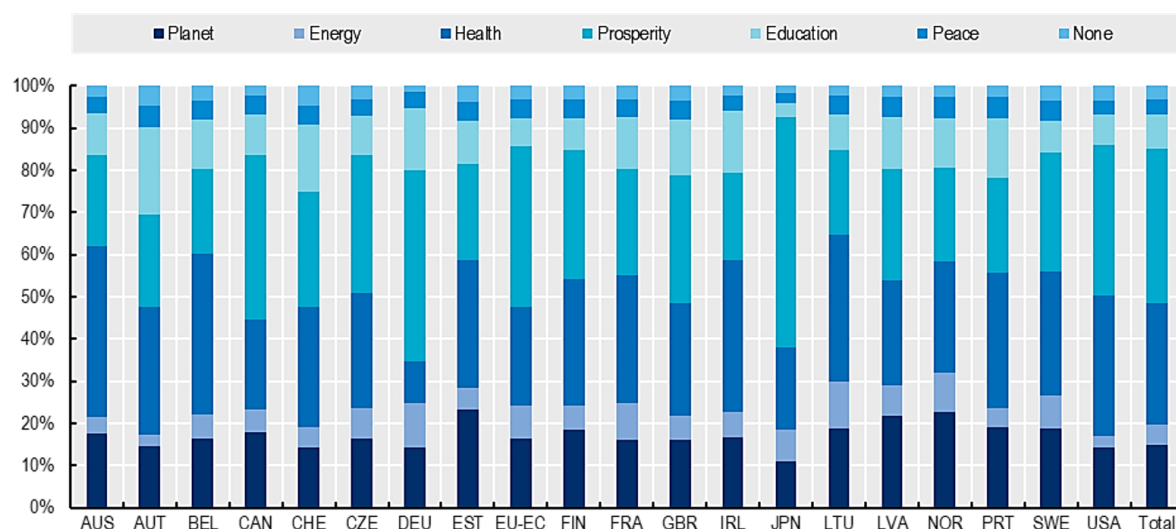


Note: Experimental estimates. Results may not be fully comparable because of limited R&D funding coverage for specific countries in Fundstat. The analysis excludes projects with missing or nil funding information, which obey disclosure controls or database entries intended to register project modifications without funding implications.

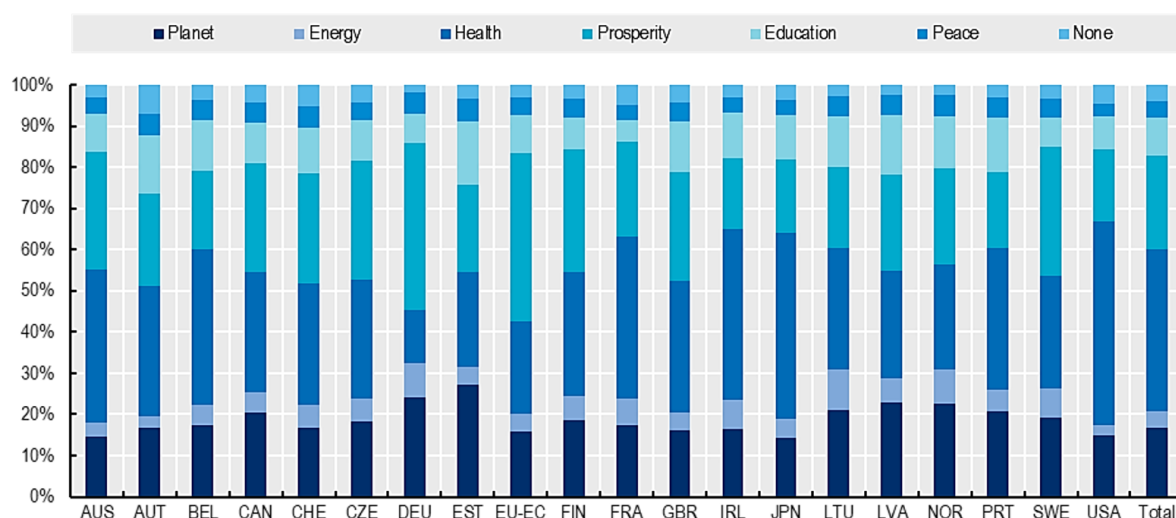
Source: OECD analysis of Fundstat database (v.2024).

Figure 5.6. SDG group distribution of R&D project funding by country/area, 2015-2023

(a) Share of R&D funding



(b) Share of R&D projects



Note: Experimental estimates. Results may not be fully comparable because of limited R&D funding coverage for specific countries in Fundstat. The analysis excludes projects with missing or nil funding information, which obey disclosure controls or database entries intended to register project modifications without funding implications. Please refer to Table 5.1 for the definition of SDG groups.

Source: OECD analysis of Fundstat database (v.2024).

5.1.5. Relevance trends for R&D funding and scientific publications

As most studies on SDG directionality for STI have been conducted on scientific publication data, it is pertinent to compare results for R&D funding and scientific publications. Table 5.1 shows the distribution of R&D projects and level of R&D funding in the OECD Fundstat database (all sources in the database), and the number of scientific publications in the Scopus database by SDG and SDG group. It is important to note that there are differences hampering effective comparisons between funding and bibliometric results, the most important of which is that Fundstat only covers a subset of OECD countries, while the publication analysis does not distinguish by country affiliation of authors. Analysis of the SDG relevance of

scientific publications at the level of countries using the SDG classifier is available online as part of the OECD Bibliometric Indicators website (OECD, 2025^[76]). One salient finding is that counting projects (awards) or scientific publications yields SDG3 as the largest individual SDG and Health as the largest SDG group with 39% of awards and 38% of publications (see also Figure 5.7). However, counting the value of funding awards yields a different picture, with SDG9 and SDG Group Prosperity as the largest in terms of funding (37%), followed by Health with 29%. This reflects significant higher average value of funding per award under SDG9, 1.5 times as large as for any other SDG.

Table 5.1. Estimates of R&D projects, R&D funding and scientific publications by SDG & SDG group

Estimates of R&D funding for 2015-2023 and scientific publications for 2015-2022

#	Label	Fundstat # R&D awards with funding info [2015-2023]	Fundstat R&D funding (USD PPP million) [2015-2023]	Fundstat average funding per award (USD PPP million) [2015-2023]	All Scopus # Scientific publications [2015-2022]
SDG					
9	9. Industry, innovation and infrastructure	310 582	446 660	1.44	3 805 844
3	3. Good health and well-being	667 734	342 183	0.51	7 542 997
4	4. Quality education	187 431	121 007	0.65	1 691 629
7	7. Affordable and clean energy	81 351	70 120	0.86	1 781 232
13	13. Climate action	96 051	64 703	0.67	1 196 128
12	12. Responsible consumption and production	83 231	62 403	0.75	1 069 786
11	11. Sustainable cities and communities	67 123	46 323	0.58	750 876
15	15. Life on land	84 142	44 810	0.69	794 655
2	2. Zero hunger	55 699	35 441	0.53	663 097
5	5. Gender equality	42 548	30 765	0.64	476 363
14	14. Life below water	43 439	30 724	0.72	489 816
17	17. Partnerships for the goals	40 856	30 164	0.71	473 294
10	10. Reduced inequalities	40 698	26 331	0.74	453 046
16	16. Peace, justice and strong institutions	38 320	24 726	0.65	421 181
8	8. Decent work and economic growth	43 653	23 093	0.65	491 365
1	1. No poverty	29 431	20 158	0.53	311 569
6	6. Clean water and sanitation	30 052	18 646	0.68	548 173
0	No SDG relevance	81 233	47 012	0.62	1 429 712
SDG group					
D	Prosperity (SDGs 8, 9, 10, 11)	462 056	542 407	1.17	5 501 131
C	Health (SDGs 1, 2, 3, 5)	795 413	428 548	0.54	8 994 026
A	Planet (SDGs 6, 12, 13, 14, 15)	336 914	221 286	0.66	4 098 558
E	Education (SDG 4)	187 431	121 007	0.65	1 691 629
B	Energy (SDG 7)	81 351	70 120	0.86	1 781 232
F	Peace (SDGs 16, 17)	79 176	54 891	0.69	894 475
G	None (SDG 0)	81 233	47 012	0.58	1 429 712
Total		2 023 574	1 485 270	0.73	24 390 764

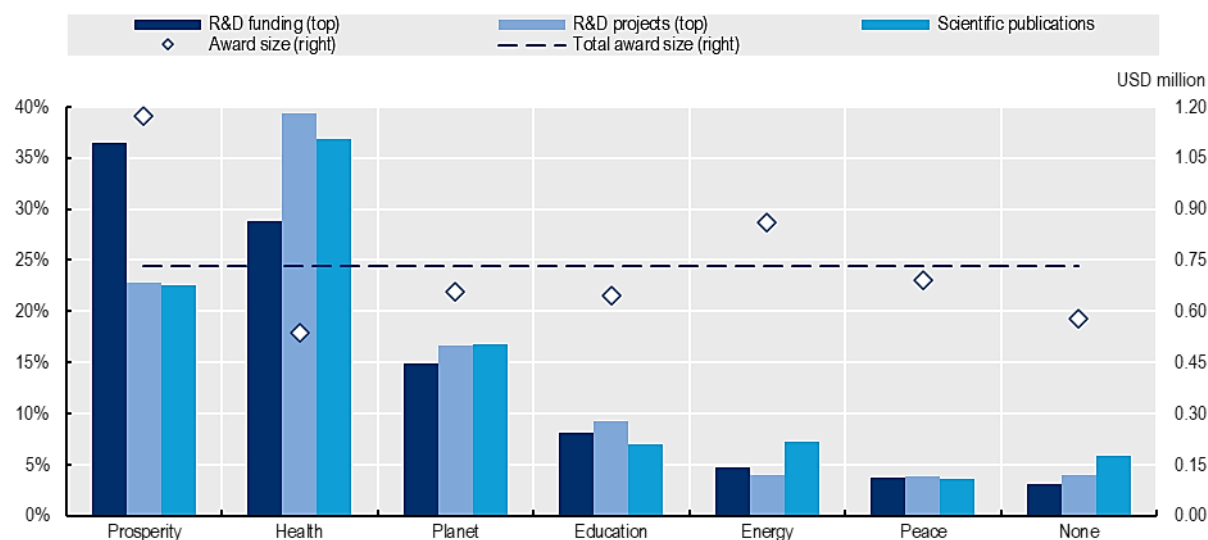
Note: Sorted by column "Fundstat R&D funding (USD PPP million)" in descending order. The calculation of award size (in USD PPP million per project) excludes projects with missing or nil funding information, which often obey to disclosure controls or database entries intended to register project modifications without funding implications (zero additional funding). The indicators on publications refer to all citeable publications in Scopus, regardless of country of author affiliation.

Source: OECD analysis of the Fundstat database (v. 2024) and Scopus database (v. January 2024).

Another salient result is that the estimate share of non-SDG relevant publications of 6% is higher than that of project awards (4%) and funding amounts (3%). Whilst this might hint that scientific publications supported by institutional funds might exhibit lower directionality towards societal goals, further work is needed to confirm that this is not driven by differences in country and thematic coverage across databases.

Figure 5.7. SDG group distribution of scientific publications and funding awards' counts and values

As a share of total Scopus database publications and R&D funding in the OECD Fundstat database



Note: The calculation of award size (in USD PPP million per project) excludes projects with missing or nil funding information, which often obey to disclosure controls or database entries intended to register project modifications without funding implications (zero additional funding). The indicators on publications refer to all citeable publications in Scopus, regardless of country of author affiliation.

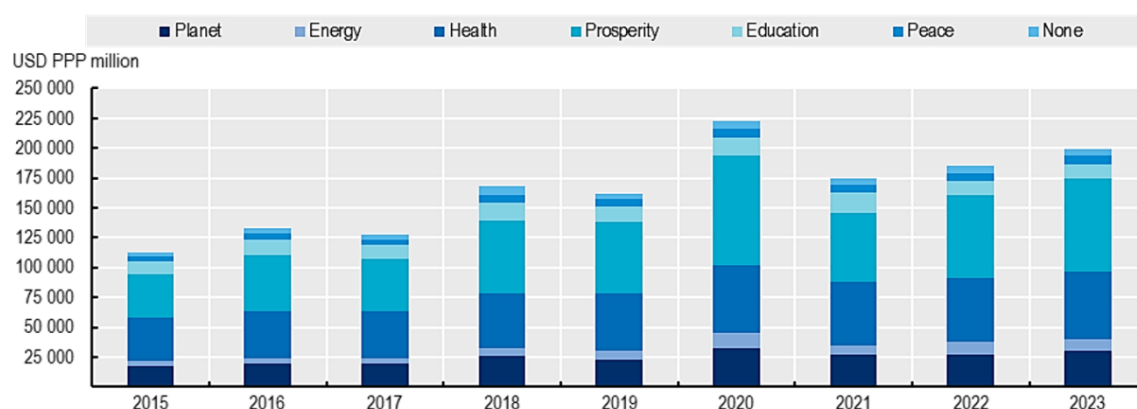
Source: OECD analysis of the Fundstat database (v. 2024) and Scopus database (v. January 2024).

Figure 5.8 and 5.9 provide more detailed trend data over the 2015-2023 period at the level of SDG groups for the three variables, in absolute terms and as shares of all possible SDGs (include no SDG option). Results at the SDG level are available in Annex C. Total R&D funding increased from ca USD 112 billion in 2015 to close to USD 200 billion in 2023, with a short-term peak of nearly USD 224 billion in 2020 most likely in response to COVID-19 crisis. The increase reported in 2020 was not principally driven by a combined surge in R&D funding directed towards SDG 9 (Industry, innovation and infrastructure) and SDG 3 (Health), more pronounced for the former. The share of R&D funding attributable to Prosperity goals increased from around 32% in 2015 to reach 39% in 2023. In contrast, total R&D project count increased up until 2020 and has declined since and was lower in 2023 than at the beginning of the period at slightly over 190 thousand awards. This appears to be related to an increased preference among funders for awards with higher levels of funding.

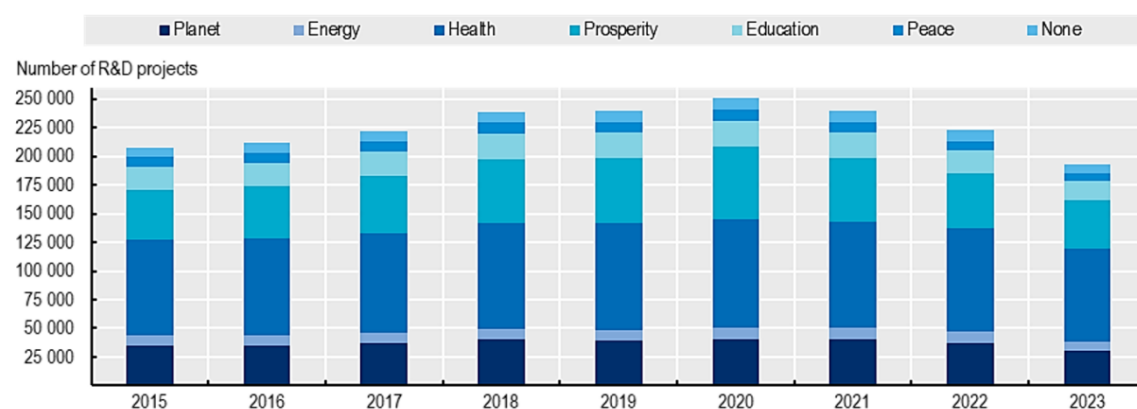
Estimates of trends in scientific publications exhibit similar behaviour to project counts as shares of total publications but exhibited continuous growth. Areas like Energy, despite exhibiting strong relative growth in publication counts (e.g., Energy publications grew by 38% over 2015-2023), remain marginal in terms of total publication share. In short, count-based indicators appear to be more stable in their goal relevance distribution than funding-based indicators regardless of whether these are input or output based. This appears to call for in depth analysis of the relationship between inputs, outputs and impacts.

Figure 5.8. Scientific publications, R&D funding and R&D projects by SDG group, 2015-2023

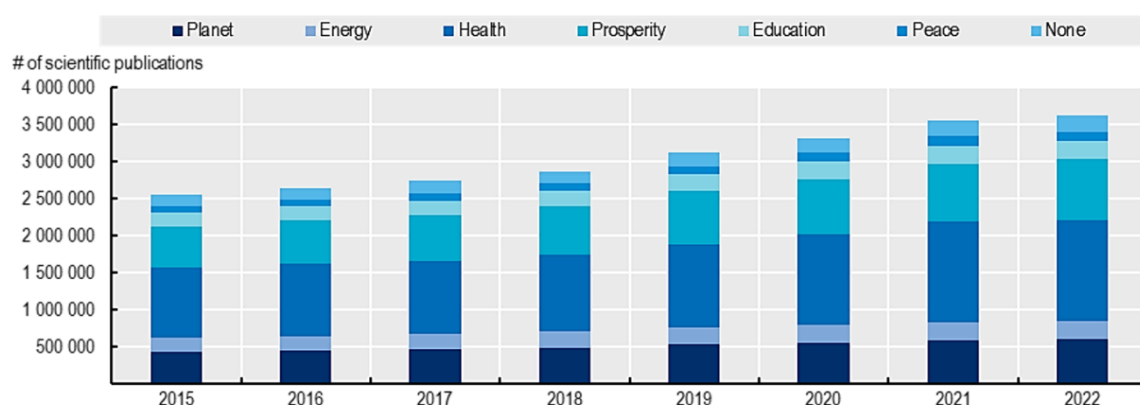
(a) Total R&D funding (USD PPP million)



(b) Total number of R&D projects



(c) Total scientific publications by SDG group (2015-22)

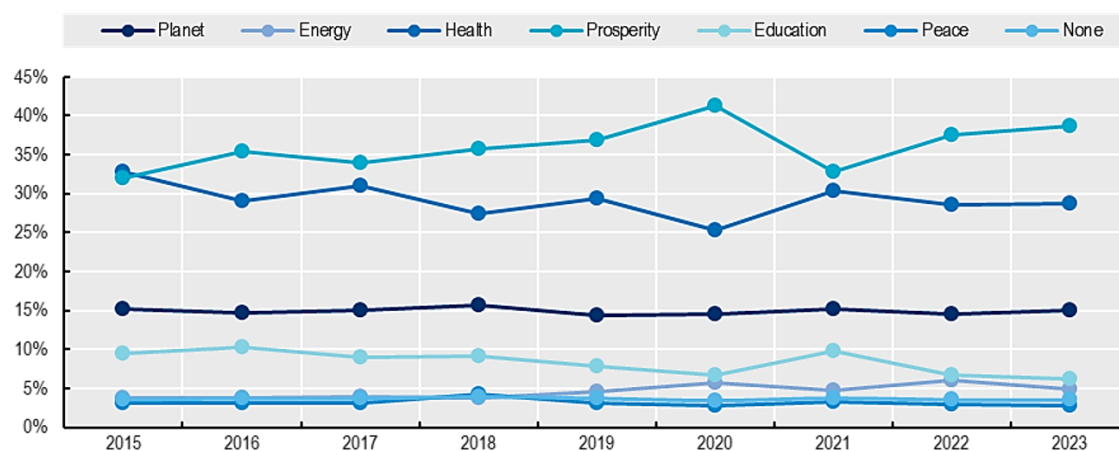


Note: The analysis excludes projects with missing or nil funding information, which obey disclosure controls or database entries intended to register project modifications without funding implications (zero additional funding).

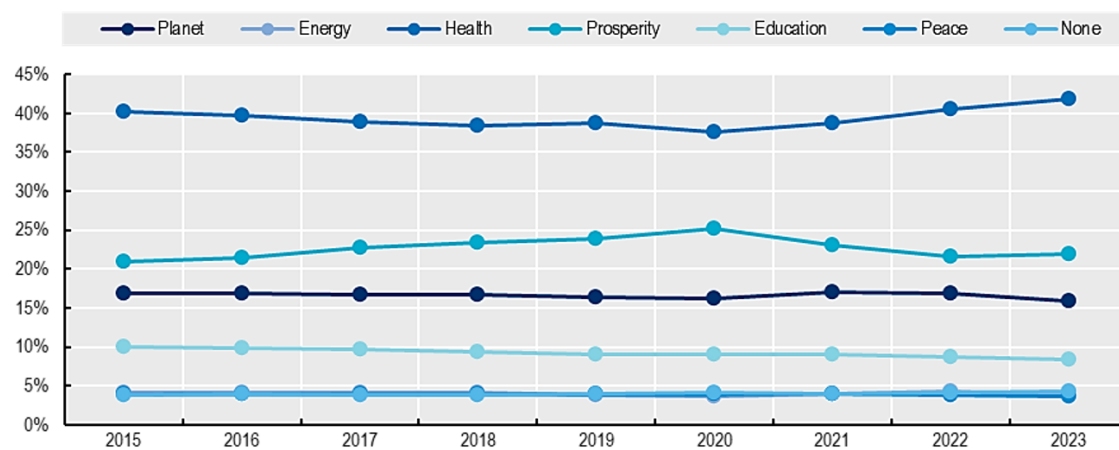
Source: OECD analysis of the Fundstat database (v. 2024) and Scopus database (v. January 2024).

Figure 5.9. Trends in shares of R&D funding, awards and publications by SDG group, 2015-2023

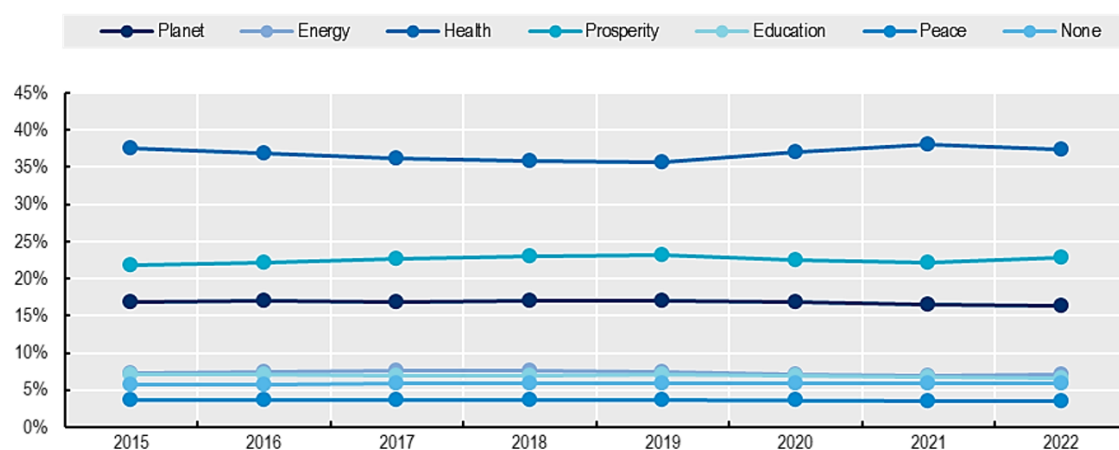
(a) R&D funding



(b) Number of R&D awards



(c) Scientific publications (2015-2022)



Note: Fractional counts for numbers of awards and scientific publications. The analysis excludes projects with missing or nil funding information, which obey disclosure controls or database entries registering project modifications without funding implications (zero additional funding).

Source: OECD analysis of the Fundstat database (v. 2024) and Scopus database (v. January 2024).

5.2. SDG R&D topics and interdependencies

5.2.1. Topic modelling of R&D projects relevant to Energy and Planet SDGs

In addition to the visualisation and analysis opportunities provided by the funding landscape analysis in the earlier section, one additional key advantage of working with granular semantic data in combination with funding values is the ability to explore detailed R&D topics within specific areas, highlighting what are the issues that R&D funders are grappling with to support. In other words, what are R&D funders supporting through projects that are relevant to specific societal goals? As a demonstration example, this subsection explores the funding awards that score highly on the probability of being relevant towards Energy and Planet SDGs.

A probabilistic topic-modelling methodology (Aristodemou et al., 2023^[3]) was applied to the relevant subset of “energy and environmental” award descriptions in the OECD Fundstat database for 2020-23. The model was calibrated to balance thematic granularity and interpretability and achieve a manageable number of topics. The analysis yielded 37 latent research topics. Each R&D project’s funding is fractionally apportioned across these 37 topics according to the model’s normalised posterior probabilities. To ensure contextually relevant descriptors, topic labels are generated and refined via OpenAI’s GPT-4.5 using the topic generated top 10 keywords and a sample of project titles and abstracts for each topic (OpenAI, 2025^[77]). For visualisation purposes, Figure 5.10 shows the result of implementing a subsequent Principal Component Analysis (PCA) on the topic probability distributions for the selected awards and plotting the topics. The PCA reduces the 37-dimensional thematic space to two orthogonal components, which combined account for 27.5% of the data variance.

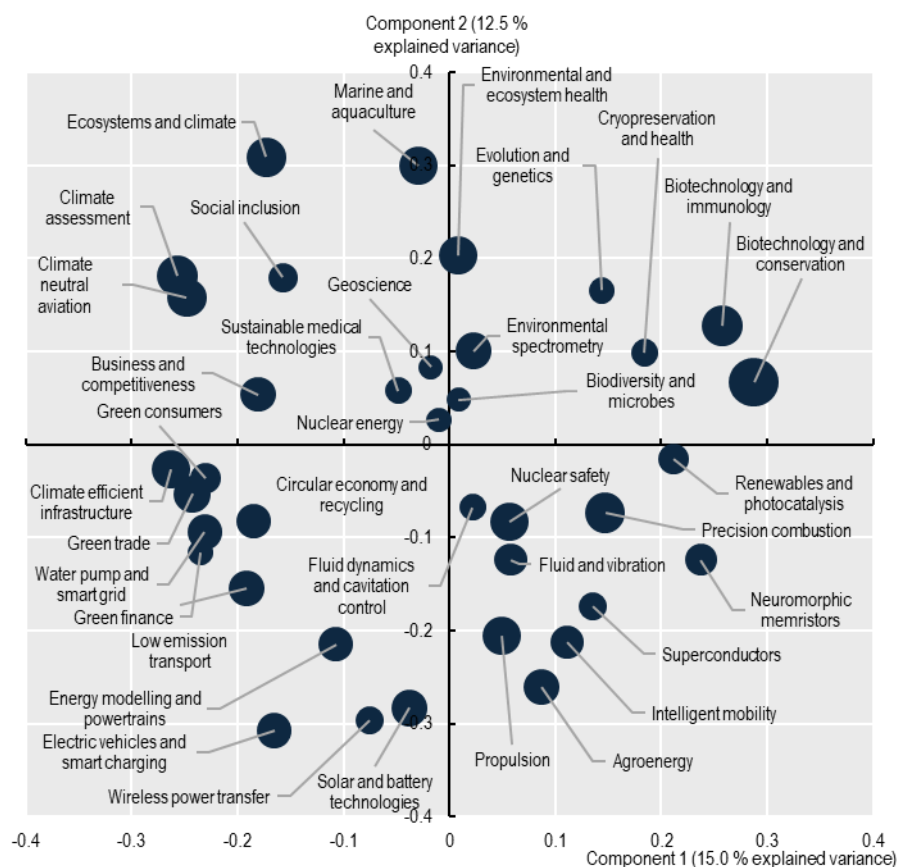
The first principal component, displayed on the horizontal axis in the biplot, explains about 15.0% of the variance in the data. Topics with high positive loadings on component 1 include biotechnology, photocatalysis, (energy efficient) neuromorphic computing and precision combustion. Negative loadings on this factor are found for topics such as Low-emission transportation, Green finance and Climate-efficient infrastructure.

While it is not straightforward to interpret PCA factors, it might appear as if Component 1 helps distinguish between topics that spell out scientific evidence or technologies that are or may be relevant for societal energy or environmental goals, and topics that directly speak to the energy and environmental challenges that R&D projects aim to address, and may touch for example on issues of technology adoption and diffusion. It is important to note that any specific project will potentially combine those different types of topics to a varying degree. This confirms that the language in project descriptions (and in the text in scientific publication abstracts) may combine elements about the substance of the R&D work and their relevance towards societal goals.

Turning to the second component (vertical axis), topics with positive loadings include Marine and aquaculture, Ecosystems and climate and Environmental and ecosystem health, whilst negative scores are found for Electric vehicles, Solar and battery technologies, Agroenergy and Propulsion. A somewhat crude interpretation would be that this dimension helps differentiate across environmental (top) and energy-(bottom) focused topics, indeed reflecting the choice made of analysing projects scoring high on relevance towards either. Again, any given project may contain a mix of both types of topics, but the emphasis will differ across the funding portfolios.

Figure 5.10. Topics in R&D projects relevant to Energy and Planet related SDGs, 2020-2023

Principal component analysis (PCA) biplot of estimated topic relevance probabilities



Note: Results of topic modelling for R&D awards with high scores on energy and environmental SDG relevance. The analysis is limited to R&D projects with available funding information. The topic model (Aristodemou et al., 2023^[3]) identifies 37 topics. Awards are probabilistically assigned to topics based on standardised probabilities predicted by the topic model. Topic labelling has been performed using OpenAI's GPT-4.5 using the topic generated top 10 keywords and a sample of project titles and abstracts for each topic. The visualisation presents the loadings for the 37 topics on the first two factors extracted via principal component analysis (PCA), depicted as the main axes. The two main principal components explain 27.5% of the variance in the data. Bubble size reflects the estimated share of total R&D funding for each topic.

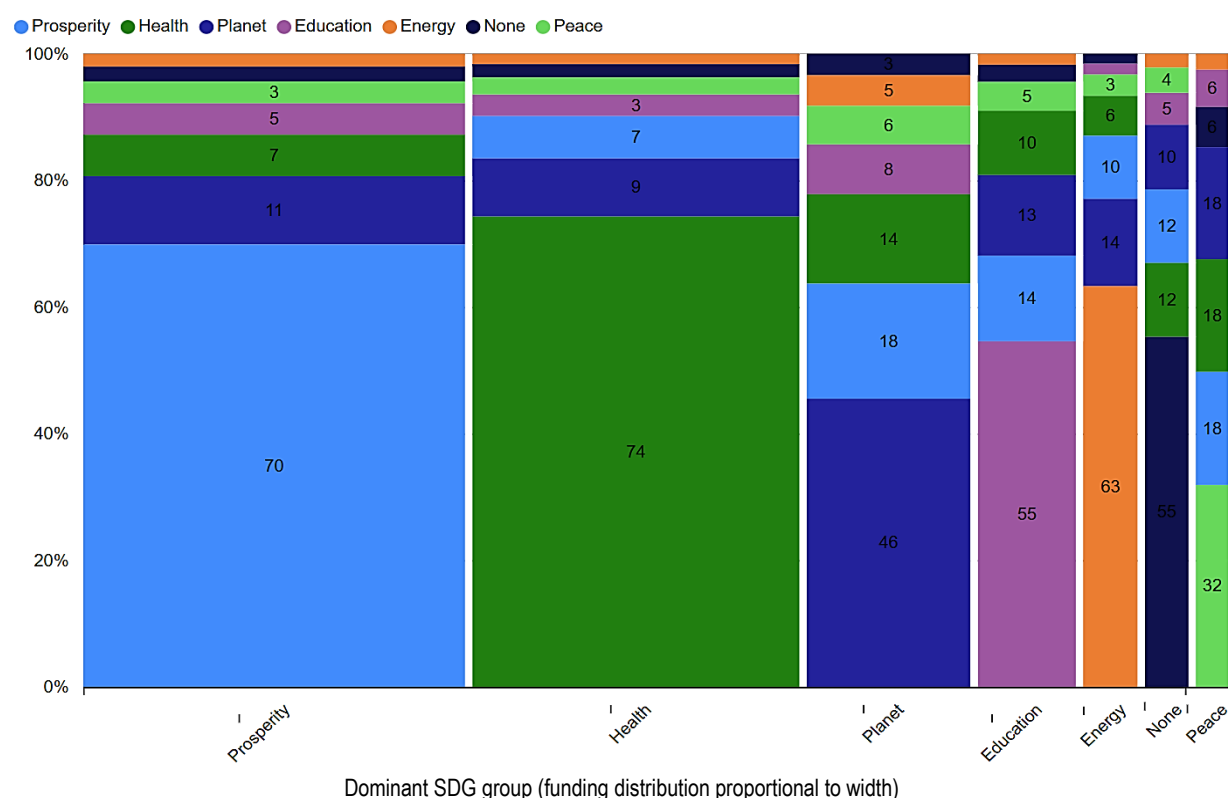
Source: OECD analysis of the OECD Fundstat database (v.2024).

5.2.2. Interdependencies in R&D funding between SDGs

SDG interdependencies may arise as R&D awards contribute towards multiple policy goals. One way to think about this is to consider how funding is distributed across projects depending on their dominant SDG group, and then within those projects across both dominant and other SDGs. Figure 5.11 visualises how the funding of R&D projects distributes across multiple SDG groups, as the areas represent the total volume of funding attributed to each dominant SDG group (columns of varying width) and, within those, how much is relevant to each SDG group (within columns with dominant SDG at the bottom).

Figure 5.11. SDG relevance of R&D funding, by dominant SDG group in R&D awards, 2010-2023

Distribution of funding relevance within projects' dominant SDG groups (horizontal), as % of dominant



Note: The Marimekko chart shows how projects principally under a SDG group contribute other secondary SDG groups. The x-axis represents different dominant (primary) SDG groups. The y-axis shows the relative distribution of secondary SDG groups within each dominant (primary) group. The analysis is limited to R&D projects with available funding information. How to read: Within "Health" dominated R&D awards, 26% (100-74%) of funding is relevant towards other SDG groups (of which 9% is Planet).

Source: OECD analysis of the OECD Fundstat database (v.2024).

The analysis shows there are substantial relevance synergies with R&D awards notionally under a dominant SDG group contributing to other SDGs. In the case of R&D projects that feature Prosperity as the dominant SDG group, nearly 20% of their R&D funding is directed towards goals such as Planet and Health. Projects predominantly assigned to Health goals, which are the least overlapping of all, are still estimated to contribute to other SDGs with 26% of their funds (9% to the SDG group Planet and 7% to Prosperity). Within funding for Planet-dominated R&D projects, 18% contributes to Prosperity and 14% to Health. 14% of Energy-dominated funding is oriented towards the SDG group Planet and 10% to Prosperity. Importantly, smaller shares of R&D funding across most (dominant) SDG groups are also connected to Peace-related goals, albeit at lower levels.

6 Concluding remarks

This report presents a novel method for attributing societal goal relevance to public investments in research and development (R&D). Using the UN SDGs as illustrative reference framework, it documents how it was possible to develop a conceptually relevant goal classifier for R&D activity descriptions that is informed by a domain-specific training database. This approach brings together the relative strengths of survey-based tools, different administrative data sources (both bibliographic and funding databases), and large language models, facilitating a balanced assessment of the relevance of R&D towards societal goals in ways that such methods, in isolation, cannot provide. While this paper's approach might overstate the effective final contribution of R&D towards the SDGs, its analysis provides a sound and plausible set of estimates of potential relevance which does not understate the foundational contribution of basic research towards addressing major socioeconomic and environmental challenges.

Whilst potentially not representative of the full breadth of governmental R&D funding across all countries in the analysis, the results provide evidence on the direction of R&D funding channelled via individually identifiable awards. The results help characterise the funding portfolios of R&D funding bodies, their implicit priorities, and provide a guide towards strategic positioning and strategy implementation.

These results on the relevance of R&D funding are also complemented by evidence from the body of scientific publications, calculated using identical methods. The additional results suggest that, contrary to previous literature, most of the scientific literature can be deemed relevant towards SDGs, and not just the fraction that includes explicit language that enables a direct connection to the SDGs as they are currently formulated.

The approaches developed in this paper also help spell out the advantages of working with statistical and administrative microdata coupled with flexible categorisation tools. The results show the potential to investigate what topics are being addressed under R&D projects relevant to specific SDGs, a particularly valuable form of timely STI intelligence since R&D funding precedes publications and other STI outputs. Furthermore, the micro-based and flexible probabilistic approach developed in this paper demonstrates synergies across different goals regardless of the need to produce meaningful additive breakdowns for funding amounts and project counts.

This study provides experimental evidence for aligning public project-based R&D funding with societal goals. By combining domain-informed researcher inputs with machine-learning classification, the study offers an informed and interpretable mapping of R&D towards societal goals. The application of the classifier to Fundstat and Scopus facilitates an assessment of how current R&D portfolios contribute to the achievement of these goals. Policymakers and funding bodies may leverage these results to identify possible R&D funding gaps and encourage R&D investments with multiple interconnected goals. Cross-country differences in the ex-ante and ex-post directionality of R&D funding towards societal goals imply opportunities for policy learning and international coordination.

There are multiple possible avenues for future work by OECD and other parties. There is indeed considerable potential to use these methods to assist in the reporting of R&D statistics and the development of relevant quality guidelines. The methods developed can add significant versatility in reporting beyond fixed high level classification schemas. All this requires the continued development of

the Fundstat data infrastructure as the most sophisticated AI-based analysis are only as good as the underlying data beneath. This requires a continued distributed effort within and across countries.

This work can also help inform the development and update of R&D classification schemas, showing inconsistencies, overlaps, blind spots and gaps. Standard R&D classifications need updating and stress testing so that they can be used with different types of data. Given the difficulties in reaching international consensus on R&D classification hierarchies for either socioeconomic objectives or fields of R&D, data-informed methods are a necessary tool to facilitate such a process.

Furthermore, there is a significant need for practical experience in making informed decisions on the potential degree of automatization in AI-based model development and inference when it comes to STI analysis. This paper has shown that expert training input was a necessary part for the classification of project descriptions to societal goals, but this does not need to be the case for other features of interest.

There are also major policy-oriented research opportunities through the examination of other dimensions of R&D activity in conjunction with societal goals. Potential areas for additional follow-on work include the analysis of Technological Readiness Levels (TRLs) in funded R&D activity, conducting deep dives on specific goals, assessing novelty in R&D funding and exploring emergence through new topic bursts.

Improving data coverage and continuing to support, grow and refine AI-based classification tools will be necessary for further advancing the existing evidence base on the directionality of R&D funding towards societal goals and making STI policy more strategic, targeted, and responsive to long-term sustainability challenges.

References

- Ambalavanan, A. and M. Devarakonda (2020), "Using the contextual language model BERT for multi-criteria classification of scientific articles", *Journal of Biomedical Informatics*, Vol. 112, <https://doi.org/10.1016/j.jbi.2020.103578>. [68]
- Aristodemou, L. (2020), *Identifying Valuable Patents: A Deep Learning Approach*, University of Cambridge, Cambridge, <https://doi.org/10.17863/CAM.69403>. [74]
- Aristodemou, L. et al. (2023), "Measuring governments' R&D funding response to COVID-19: An application of the OECD Fundstat infrastructure to the analysis of R&D directionality", *OECD Science, Technology and Industry Working Papers*, No. 2023/06, OECD Publishing, Paris, <https://doi.org/10.1787/4889f5f2-en>. [3]
- Armitage, C., M. Lorenz and S. Mikki (2020), "Mapping scholarly publications related to the Sustainable Development Goals: Do independent bibliometric approaches get the same results?", *Quantitative Science Studies*, Vol. 1/3, pp. 1092-1108, https://doi.org/10.1162/qss_a_00071. [32]
- Bautista-Puig, N. et al. (2021), "Unveiling the Research Landscape of Sustainable Development Goals and Their Inclusion in Higher Education Institutions and Research Centers: Major Trends in 2000–2017", *Frontiers in Sustainability*, Vol. 2, <https://doi.org/10.3389/frsus.2021.620743>. [34]
- Bedard-Vallee Alexandre, James Chris and Roberge Guillaume (2023), "2023 Sustainable Development Goals (SDGs) now available on Scopus", *Elsevier Data Repository*, <https://doi.org/10.17632/y2zyy9vwzy.1>. [42]
- Bello, M. and F. Galindo-Rueda (2020), "Charting the digital transformation of science: Findings from the 2018 OECD International Survey of Scientific Authors (ISSA2)", *OECD Science, Technology and Industry Working Papers*, No. 2020/03, OECD Publishing, Paris, <https://doi.org/10.1787/1b06c47c-en>. [53]
- Beltagy, I., K. Lo and A. Cohan (2019), *SCIBERT: A pretrained language model for scientific text*, <https://doi.org/10.18653/v1/d19-1371>. [61]
- Borchardt, S. et al. (2023), *Mapping EU policies with the 2030 agenda and SDGs: fostering policy coherence through text-based SDG mapping*, Publications Office of the European Union, <https://doi.org/10.2760/110687>. [52]

- Boudt, K., Y. Inghels and A. Spithoven (2024), *Trac(k)ing the SDG 8 trajectory Trac(k)ing the Trajectory: Mapping Sustainable Development Goal 8 in EU-Funded Research Projects*, Elsevier BV, <https://doi.org/10.2139/ssrn.4898389>. [36]
- Church, K., Z. Chen and Y. Ma (2021), "Emerging trends: A gentle introduction to fine-tuning", *Natural Language Engineering*, Vol. 27/6, pp. 763-778, <https://doi.org/10.1017/s1351324921000322>. [62]
- Confraria, H., T. Ciarli and E. Noyons (2024), "Countries' research priorities in relation to the Sustainable Development Goals", *Research Policy*, Vol. 53/3, p. 104950, <https://doi.org/10.1016/j.respol.2023.104950>. [27]
- Cui, Y. et al. (2019), "Class-Balanced Loss Based on Effective Number of Samples", *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 9260-9269, <https://doi.org/10.1109/cvpr.2019.00949>. [70]
- Dan Clark et al. (2023), *Artificial Intelligence: Generative AI exists because of the transformer (by the visual storytelling team)*, Financial-Times, <https://ig.ft.com/generative-ai/> (accessed on 26 September 2024). [59]
- Devlin, J. et al. (2019), *BERT: Pre-training of deep bidirectional transformers for language understanding*, <https://doi.org/10.48550/arXiv.1810.04805>. [67]
- Elsevier (2022), *Times Higher Education Impact Rankings, Scopus and SciVal*, <https://www.elsevier.com/academic-and-government/impact-rankings-data> (accessed on 25 September 2024). [44]
- Hajikhani, A. and A. Suominen (2022), "Mapping the sustainable development goals (SDGs) in science, technology and innovation: application of machine learning in SDG-oriented artefact detection", *Scientometrics*, Vol. 127/11, pp. 6661-6693, <https://doi.org/10.1007/s11192-022-04358-x>. [30]
- Hu, Q. et al. (2025), "A weighted difference loss approach for enhancing multi-label classification", *Scientific Reports*, Vol. 15/1, <https://doi.org/10.1038/s41598-025-09883-2>. [71]
- IEA (2021), *Net Zero by 2050*, IEA, Paris, <https://www.iea.org/reports/net-zero-by-2050> (accessed on 12 September 2024). [51]
- IPCC (2022), *Climate Change 2022 – Impacts, Adaptation and Vulnerability*, Cambridge University Press, <https://doi.org/10.1017/9781009325844>. [50]
- IPCC (2021), *Climate Change 2021 – The Physical Science Basis*, Cambridge University Press, <https://doi.org/10.1017/9781009157896>. [49]
- Johnson, J. and T. Khoshgoftaar (2019), "Survey on deep learning with class imbalance", *Journal of Big Data*, Vol. 6/1, <https://doi.org/10.1186/s40537-019-0192-5>. [69]
- Jurafsky, D. and J. Martin (2023), *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition*, Pearson Education, <https://web.stanford.edu/~jurafsky/slp3/> (accessed on 1 May 2023). [60]
- Kashnitsky, Y. et al. (2024), "Evaluating approaches to identifying research supporting the United Nations Sustainable Development Goals", *Quantitative Science Studies*, Vol. 5/2, pp. 408-425, https://doi.org/10.1162/qss_a_00304. [8]

- LaFleur, M. (2023), *Using Large Language Models to Help Train Machine Learning SDG Classifiers*, United Nations (UN) Department of Economic and Social Affairs (DESA), <https://doi.org/10.18356/25206656-180>. [38]
- LeCun, Y., Y. Bengio and G. Hinton (2015), "Deep learning", *Nature*, Vol. 521/7553, pp. 436-444, <https://doi.org/10.1038/nature14539>. [73]
- Lecun, Y. et al. (1998), "Gradient-based learning applied to document recognition", *Proceedings of the IEEE*, Vol. 86/11, pp. 2278-2324, <https://doi.org/10.1109/5.726791>. [66]
- Lin, T. et al. (2017), "Focal Loss for Dense Object Detection", *2017 IEEE International Conference on Computer Vision (ICCV)*, <https://doi.org/10.1109/iccv.2017.324>. [72]
- Massucci, F. and A. Seri (2022), *Exploratory study understanding the SDG alignment along research activities and technological innovation of scoreboard companies*, Joint Research Centre (JRC), Publications Office of the European Union, <https://doi.org/10.2760/51016>. [28]
- OECD (2025), "Analysis of the SDG directionality of scientific publications at the level of countries", *OECD Science and bibliometric indicators*, <https://www.oecd.org/en/data/datasets/science-and-bibliometric-indicators.html> (accessed on 4 September 2025). [76]
- OECD (2025), "Indicator (C_INDKNOWXGB) on R&D budget for Industry and Knowledge (SDG) as a percentage of total Government support", *OECD calculations based on OECD, Research and Development Statistics (RDS)*, <http://oe.cd/rds>, and *OECD, R&D Tax Incentive*, <http://oe.cd/rdtax>, Databases, https://stip.oecd.org/stats/SB-StatTrends.html?i=C_INDKNOWXGB (accessed on 2 September 2025). [23]
- OECD (2025), "Indicator (C_PLANXGB) on R&D budget for Planet and Infrastructure (SDG) as a percentage of total Government support", *OECD calculations based on OECD, Research and Development Statistics (RDS)*, <http://oe.cd/rds>, and *OECD, R&D Tax Incentive*, <http://oe.cd/rdtax>, Databases, https://stip.oecd.org/stats/SB-StatTrends.html?i=C_PLANXGB (accessed on 2 September 2025). [24]
- OECD (2025), "Indicator (C_SANTSOCXGB) on R&D budget for Health and Society (SDG) as a percentage of total Government support", *OECD calculations based on OECD, Research and Development Statistics (RDS)*, <http://oe.cd/rds>, and *OECD, R&D Tax Incentive*, <http://oe.cd/rdtax>, Databases, https://stip.oecd.org/stats/SB-StatTrends.html?i=C_SANTSOCXGB (accessed on 2 September 2025). [22]
- OECD (2025), "Indicator (C_SECXGB) on R&D budget for Security (SDG) as a percentage of total Government support", *OECD calculations based on OECD, Research and Development Statistics (RDS)*, <http://oe.cd/rds>, and *OECD, R&D Tax Incentive*, <http://oe.cd/rdtax>, Databases, https://stip.oecd.org/stats/SB-StatTrends.html?i=C_SECXGB (accessed on 2 September 2025). [25]
- OECD (2025), "Main Science and Technology Indicators (MSTI)", *OECD Science, Technology and R&D Statistics*, <https://www.oecd.org/en/data/datasets/main-science-and-technology-indicators.html> (accessed on 2 May 2025). [57]

- OECD (2025), *Methodology note: Measuring Distance to SDGs*, [20]
https://www.oecd.org/content/dam/oecd/en/data/dashboards/the-oecd-hub-on-measuring-distance-to-sdg-targets/OECD%20SDG%20Hub%20Methodology%20Note.pdf/jcr_content/renditions/original/OECD%20SDG%20Hub%20Methodology%20Note.pdf.
- OECD (2025), “R&D spending growth slows in OECD, surges in China; government support for energy and defence R&D rises sharply”, *Main Science and Technology Indicators, OECD Science, Technology and R&D Statistics*, <https://www.oecd.org/en/data/insights/statistical-releases/2025/03/rd-spending-growth-slows-in-oecd-surges-in-china-government-support-for-energy-and-defence-rd-rises-sharply.html> (accessed on 2 May 2025). [58]
- OECD (2025), “Science, Technology and Innovation Scoreboard”, *OECD Science, Technology and R&D Statistics*, <https://www.oecd.org/en/data/datasets/science-technology-and-innovation-scoreboard.html> (accessed on 2 May 2025). [26]
- OECD (2025), *The OECD Hub on Sustainable Development Goals*, [21]
<https://www.oecd.org/en/data/dashboards/the-oecd-hub-on-sustainable-development-goals.html> (accessed on 20 June 2025).
- OECD (2024), “Declaration on Transformative Science, Technology and Innovation Policies for a Sustainable and Inclusive Future”, *OECD/LEGA/0501*,
<https://legalinstruments.oecd.org/en/instruments/OECD-LEGAL-0501> (accessed on 10 September 2024). [5]
- OECD (2024), *How do governments direct support for innovation? Lessons from recent OECD measurement and impact analysis (MABIS) work*, OECD Publishing, Paris,
<https://doi.org/10.1787/c1d93d1c-en>. [11]
- OECD (2024), “How to direct research and innovation funding and finance to address the climate challenge?”, *Shared challenges, transformative actions, OECD Science and Technology Policy Ministerial*, OECD Publishing, <https://cdn-assets.inwink.com/693d4b04-1de5-4c06-808d-2f0c6b423782/6aa8eaf8-0fe5-471a-914f-f0ef810635ab?sv=2018-03-28&sr=b&sig=yryj3srzMRsviZhwPnpbcw%2FeEyUEe1Z4tpasPsvunJ4%3D&se=9999-12-31T23%3A59%3A59Z&sp=r&rscd=inline%3B%20filename%3D%22MIN24-Key%2520Issues%2520Note%2520BRB%2520%28rev1%29.pdf%22> (accessed on 30 September 2024). [7]
- OECD (2024), *International Survey of Science (ISSA), 2024*, [54]
<https://www.oecd.org/en/data/datasets/international-survey-of-science.html> (accessed on 17 September 2024).
- OECD (2024), “OECD Agenda for Transformative Science, Technology and Innovation Policies”, *OECD Science, Technology and Industry Policy Papers*, No. 164, OECD Publishing, Paris,
<https://doi.org/10.1787/ba2aaf7b-en>. [6]
- OECD (2023), *Government support for business innovation: Results from measurement pilots in five OECD countries*, [https://one.oecd.org/document/DSTI/STP/NESTI\(2023\)3/FINAL/en/pdf](https://one.oecd.org/document/DSTI/STP/NESTI(2023)3/FINAL/en/pdf). [9]
- OECD (2023), “OECD framework for mapping and quantifying government support for business innovation”, *OECD Science, Technology and Industry Policy Papers*, No. 160, OECD Publishing, Paris, <https://doi.org/10.1787/f30547e7-en>. [10]

- OECD (2023), "The Impact of R&D tax incentives: Results from the OECD microBeRD+ project", *OECD Science, Technology and Industry Policy Papers*, No. 159, OECD Publishing, Paris, <https://doi.org/10.1787/1937ac6b-en>. [13]
- OECD (2020), "The effects of R&D tax incentives and their role in the innovation policy mix: Findings from the OECD microBeRD project, 2016-19", *OECD Science, Technology and Industry Policy Papers*, No. 92, OECD Publishing, Paris, <https://doi.org/10.1787/65234003-en>. [14]
- OECD (2019), *Measuring Distance to the SDG Targets 2019: An Assessment of Where OECD Countries Stand*, OECD Publishing, Paris, <https://doi.org/10.1787/a8caf3fa-en>. [19]
- OECD (2015), "Daejeon Declaration on Science, Technology and Innovation Policies for the Global and Digital Age", *OECD/LEGAL/0416*, <https://legalinstruments.oecd.org/en/instruments/OECD-LEGAL-0416> (accessed on 6 September 2024). [4]
- OECD (2015), *Frascati Manual 2015: Guidelines for Collecting and Reporting Data on Research and Experimental Development*, The Measurement of Scientific, Technological and Innovation Activities, OECD Publishing, Paris, <https://doi.org/10.1787/9789264239012-en>. [1]
- OECD (forthcoming), *Science in society in times of crisis: findings from the OECD 2021 International Survey of Science (ISSA)*, OECD Publishing, Paris. [55]
- OECD/Eurostat (2018), *Oslo Manual 2018: Guidelines for Collecting, Reporting and Using Data on Innovation, 4th Edition*, The Measurement of Scientific, Technological and Innovation Activities, OECD Publishing, Paris/Eurostat, Luxembourg, <https://doi.org/10.1787/9789264304604-en>. [12]
- OpenAI (2025), *Introducing GPT-4.5*, OpenAI, https://openai.com/index/introducing-gpt-4-5/?utm_source=chatgpt.com. [77]
- OSDG; UNDP IICPSD SDG AI Lab; PPMI (2024), *OSDG Community Dataset (OSDG-CD) (2024.04) [Data set]*, <https://doi.org/10.5281/zenodo.11441197>. [56]
- Ottaviani, M. and S. Stahlschmidt (2024), "On the performativity of SDG classifications in large bibliometric databases", <https://doi.org/10.48550/arXiv.2405.03007>. [39]
- Pan, S. and Q. Yang (2010), "A Survey on Transfer Learning", *IEEE Transactions on Knowledge and Data Engineering*, Vol. 22/10, pp. 1345-1359, <https://doi.org/10.1109/tkde.2009.191>. [64]
- Priem, J., H. Piwowar and R. Orr (2022), "OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts", *ArXiv*, <https://arxiv.org/abs/2205.01833> (accessed on 17 September 2024). [75]
- Pukelis, L. et al. (2022), "OSDG 2.0: a multilingual tool for classifying text data by UN Sustainable Development Goals (SDGs)", <https://arxiv.org/abs/2211.11252v1> (accessed on 11 September 2024). [45]
- Purnell, P. (2022), "A comparison of different methods of identifying publications related to the United Nations Sustainable Development Goals: Case study of SDG 13—Climate Action", *Quantitative Science Studies*, Vol. 3/4, pp. 976-1002, https://doi.org/10.1162/qss_a_00215. [33]

- Requejo-Castro, D., R. Giné-Garriga and A. Pérez-Foguet (2020), “Data-driven Bayesian network modelling to explore the relationships between SDG 6 and the 2030 Agenda”, *Science of The Total Environment*, Vol. 710, p. 136014, <https://doi.org/10.1016/j.scitotenv.2019.136014>. [47]
- Roberge, G. and C. James (2025), “Elsevier 2025 Sustainable Development Goals (SDGs) Mapping”, *Elsevier Data Repository*, V1, <https://doi.org/10.17632/p52c7d3hfs.1>. [43]
- Romero Goyeneche, O. et al. (2022), “Mobilizing the transformative power of research for achieving the Sustainable Development Goals”, *Research Policy*, Vol. 51/10, p. 104589, <https://doi.org/10.1016/j.respol.2022.104589>. [35]
- Roy, J. et al. (2018), “Sustainable Development, Poverty Eradication and Reducing Inequalities”, in Masson-Delmotte, V. (ed.), *Global Warming of 1.5°C. An IPCC Special Report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change*, Cambridge University Press, <https://doi.org/10.1017/9781009157940.007>. [48]
- Schmidt, F. and M. Vanderfeesten (2021), “Evaluation on accuracy of mapping science to the United Nations’ sustainable development goals (SDGs) of the Aurora SDG queries”. [41]
- UN (2024), *SDG Indicators Metadata repository*, United Nations, Department of Economic and Social Affairs (DESA), Statistics Division, <https://unstats.un.org/sdgs/metadata/> (accessed on 27 September 2024). [18]
- UN (2024), *The 17 Sustainable Development Goals (SDG)*, United Nations, Department of Economic and Social Affairs (DESA), <https://sdgs.un.org/goals> (accessed on 18 September 2024). [16]
- UN (2024), *UN Data Commons for the SDGs*, United Nations, Department of Economic and Social Affairs (DESA), Statistics Division, <https://unstats.un.org/UNSDWebsite/undatacommons/sdgs> (accessed on 27 September 2024). [17]
- UN (2017), *71/313. Work of the Statistical Commission pertaining to the 2030 Agenda for Sustainable Development*, United Nations General Assembly Resolution, <https://docs.un.org/en/A/RES/71/313>. [15]
- Vanderfeesten, M., R. Jaworek and L. Keßler (2022), “AI for mapping multi-lingual academic papers to the United Nations’ Sustainable Development Goals (SDGs)”, <https://doi.org/10.5281/zenodo.6487606>. [40]
- Vaswani, A. et al. (2017), *Attention is all you need*, <https://doi.org/10.48550/arXiv.1706.03762>. [65]
- Wang, J. and Y. Chen (2022), “Pre-Training and Fine-Tuning”, in *Machine Learning: Foundations, Methodologies, and Applications, Introduction to Transfer Learning*, Springer Nature Singapore, Singapore, https://doi.org/10.1007/978-981-19-7584-4_8. [63]
- Wincott, C. (2024), “SDGs: how do government priorities and research interests align?”, *Overton blog - Impact tracking*, <https://www.overton.io/blog/sdgs-how-do-government-priorities-and-research-interests-align> (accessed on 10 June 2025). [31]

- WIPO (2024), “Mapping Innovations: Patents and the Sustainable Development Goals”, *WIPO Patent Analytics*, <https://www.wipo.int/en/web/patent-analytics/mapping-innovations-patents-sustainable-development-goals> (accessed on 10 June 2025). [29]
- Wulff, D., D. Meier and R. Mata (2024), “Using novel data and ensemble models to improve automated labeling of Sustainable Development Goals”, *Sustainability Science*, Vol. 19/5, pp. 1773-1787, <https://doi.org/10.1007/s11625-024-01516-3>. [37]
- Yamashita, I. et al. (2021), “Measuring the AI content of government-funded R&D projects: A proof of concept for the OECD Fundstat initiative”, *OECD Science, Technology and Industry Working Papers*, No. 2021/09, OECD Publishing, Paris, <https://doi.org/10.1787/7b43b038-en>. [2]
- Yin, H. et al. (2023), “Leveraging Artificial Intelligence Technology for Mapping Research to Sustainable Development Goals: A Case Study”, <https://doi.org/10.48550/arXiv.2311.16162>. [46]

Annex A. The OECD Fundstat database

A.1. Data sources in the OECD Fundstat database

Table A.1 provides an overview of the government R&D and innovation funding agencies with data sources in each country, in the 19 OECD countries and the European Union (EU) – European Commission (EC) in the OECD Fundstat database (v.2024). For 2015-2023, Fundstat contains approximately 2.1 million R&D project awards, worth approximately 1.5 billion USD PPP. Less than 1% (19 385) of the R&D project awards have no textual information available, in either the title or the description. About 6% (124 441) of the R&D project awards have no available funding information.

Table A.1. Data sources in the OECD Fundstat database, 2015-2023

ISO code	Country	Government R&D and Innovation Funding Organisation / Data Source	Label used for abbreviation	R&D project awards	R&D funding (USD PPP million)
AUS	Australia	Australian Research Council	AUS_ARC	10 641	5 141
		Australian Renewable Energy Agency	AUS_ARENA	399	1 014
		Department of Health and Aged Care	AUS_DHAC	1 391	3 036
		GrantConnect	AUS_GC	15 881	8 390
		National Health and Medical Research Council	AUS_NHMRC	9 419	5 949
AUT	Austria	Federal Research Database	AUT_BFDAT	2 177	3 460
		Federal Ministry of Agriculture	AUT_BML	1 173	n/a
		Austrian Science Fund	AUT_FWF	6 618	3 120
BEL	Belgium	Flanders Research Information Space	BEL_FRIS	43 220	6 135
		Belgian Science Policy Office INVENT database	BEL_INVENT	15 582	2 034
CAN	Canada	Canadian Institute of Health Research	CAN_CIHR	26 388	2 098
		Contracts	CAN_CON	972	63
		Grant Contributions	CAN_GC	54 707	40 372
		Natural Resources Canada	CAN_NRCan	68	71
		Natural Sciences and Engineering Research Council	CAN_NSERC	197 700	8 354
		Social Sciences and Engineering Research Council	CAN_SSHRC	75 029	4 820
CHE	Switzerland	Administration Research Actions Management Information Systems	CHE_ARAMIS	14 212	7 727
	Switzerland	Swiss National Science Foundation	CHE_SNSF	23 413	7 985
CZE	Czechia	Technology Agency Database	CZE_STARFOS	16 287	15 836
DEU	Germany	Federal Ministry of Education and Research	DEU_FOEKAT	80 418	85 059
EST	Estonia	Estonian Research Information System	EST_ETIS	5 385	1 562
EU-EC	European Union - European Commission	Funding and Tenders Portal	EU-EC_FTP	5 871	16 192
		CORDIS - H2020	EU-EC_H2020	34 997	99 576
		CORDIS - HORIZON	EU-EC_HORIZON	8 505	39 804
		Kohesio	EU-EC_Kohesio	109 723	52 754
FIN	Finland	Business Finland	FIN_BF	1 347	642
		Research Council Finland	FIN_RCF	3 856	1 821
FRA	France	French Environment and Energy Management Agency	FRA_ADEME	5 995	4 821
		National Research Agency	FRA_ANR	16 267	20 056
		French Agency for Food, Environmental and Occupational Health	FRA_ANSES	314	74

ISO code	Country	Government R&D and Innovation Funding Organisation / Data Source	Label used for abbreviation	R&D project awards	R&D funding (USD PPP million)
		and Safety			
		Ministry of Health	FRA_HEALTH	2 782	1 733
		Ministry of Health - French National Cancer Institute	FRA_INCA	316	186
GBR	United Kingdom	Government Grants Information System	GBR_GGIS	5 023	26 094
		National Institute for Health Research	GBR_NIHR	4 951	5 849
		United Kingdom Research and Innovation Gateway-to-Research	GBR_UKRI	94 354	53 907
IRL	Ireland	Department of Agriculture, Food and the Marine	IRL_DAFM	141	98
		Health Research Board	IRL_HBR	789	422
		Irish Research Council	IRL_IRC	4 572	-
		Research Ireland	IRL_RI	3 127	2 034
		Sustainable Energy Authority of Ireland	IRL_SEAI	357	172
JPN	Japan	Agency for Medical Research and Development	JPN_AMED	16 894	7 940
		Japan Science and Technology Agency	JPN_JST	6 222	-
		Database of Grants-in-Aid for Scientific Research	JPN_KAKEN	271 643	21 170
		New Energy and Industrial Technology Development Organisation	JPN_METI_NEDO	130	62 177
LTU	Lithuania	OpenData	LTU_OpenData	107	11
		Research Council Lithuania	LTU_SPEKTRAS	757	284
LVA	Latvia	Latvian Council of Science	LVA_NZDIS	10 325	3 575
NOR	Norway	Research Council Norway	NOR_RCN	16 845	9 306
PRT	Portugal	National Register of Science and Technology Funding	PRT_SciProj	27 498	4 675
SWE	Sweden	Swedish Current Research Information System	SWE_SweCRIS	36 738	13 856
USA	United States	Department of Energy	USA_DOE	2 825	7 461
		National Institute of Health	USA_NIH	605 341	336 214
		National Science Foundation	USA_NSF	112 334	64 518
		Federal Procurement Data System	USA_Procurement	96 832	344 710
		OtherGrants	USA_OtherGrants	38 818	81 902
Total				2 147 676	1 496 262

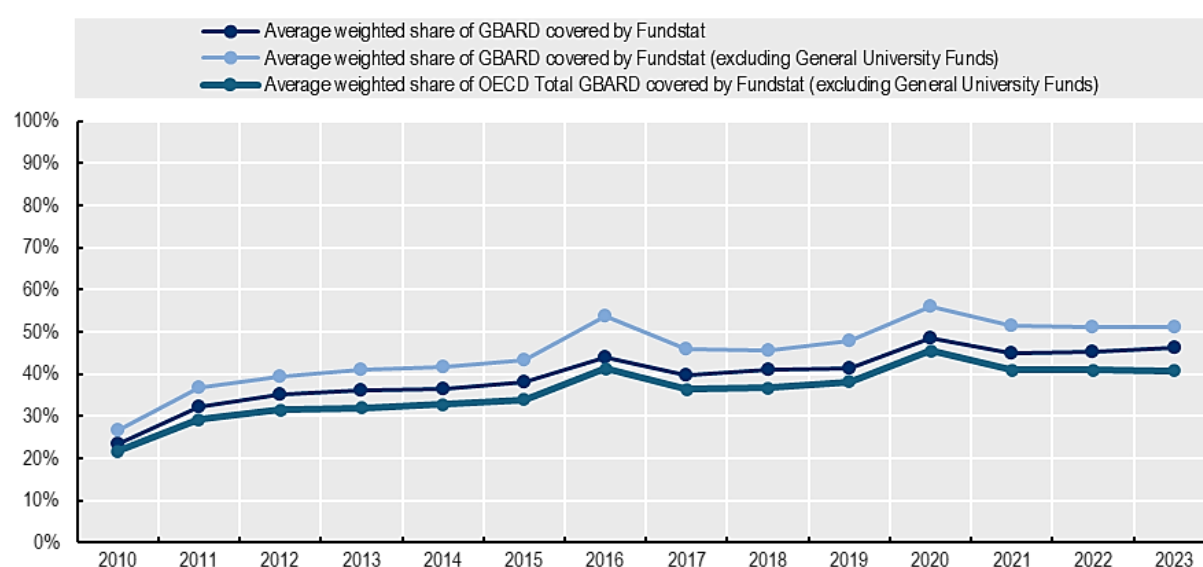
Note: Ordered in alphabetical order of column “Country/area”, followed by “Label used for abbreviation”. Each data source consists of multiple sub data sources. For example, UKRI’s Gateway-to-Research (GtR) includes Innovate UK, ESRC, NERC, EPSRC, BBSRC, MRC, AHRC, STFC, NC3Rs, UKRI etc. The total R&D funding project awards are reported regardless of the available funding or descriptive information. For Belgium (BEL), data on total R&D projects and funding is provided by the Belgian Science Policy Office (BELSPO), and by the Department of Economy, Science, and Innovation (EWI) of the Flemish government. For Japan (JPN), the project counts for JPN_METI_NEDO may reflect programmes rather than award-level projects, due to linguistic differences in translation from Japanese to English.

Source: OECD analysis of the Fundstat database (v.2024).

A.2. Benchmark analysis of the OECD Fundstat database

Over different cohorts since the start of the OECD Fundstat initiative, its working database has progressively increased the number of sources and countries covered, enhancing its coverage of national government R&D funding as well. The latest available version of the OECD Fundstat database (2024 version) displays a steady improvement in the coverage of project-level R&D funding data over the period 2010-2023. As illustrated in Figure A.1, the weighted average coverage according to each country's Government Budget Allocations for R&D (GBARD) relative to the total GBARD for the 19 OECD countries in the OECD Fundstat database, excluding General University Funds (GUF), increases from approximately 27% in 2010 to nearly 51% by 2023. Similarly, when measured as a percentage of the OECD total GBARD (excluding GUF), coverage rose from around 22% to nearly 41% over the same period. Improvements in the availability and accessibility of administrative data from national funding agencies are likely reasons, potentially coupled with changes in funding structures that may be reducing the role of institutional funding.

Figure A.1. Coverage of government R&D funding in the 2024 OECD Fundstat database, 2010-2023



Note: Percentage coverage of R&D funding in the OECD Fundstat database by country quantifies the proportion of R&D funding accessible within that country's R&D data sources (Table A.1) in the form of R&D project awards, relative to the country's Government Budget Allocations for R&D (GBARD), excluding the General University Funds (GUF). The OECD Fundstat database contains R&D funding project award data from 19 OECD countries (AUS, AUT, BEL, CAN, CHE, CZE, DEU, EST, FIN, FRA, GBR, IRL, JPN, LTV, LTU, NOR, POR, SWE, USA) and the European Union-European Commission's programmes. Government Budget Allocations for R&D (GBARD) provisional and estimated data exist until 2024 for some countries (OECD, 2025^[57]; OECD, 2025^[58]).

Source: OECD benchmark analysis of the OECD Fundstat database (v.2024).

A.3. Comparing the thematic orientation of R&D funding in Fundstat and GBARD

Whilst Fundstat allows for significant granularity in its analysis of government R&D funding, the resulting analysis may not necessarily be representative of total funding. Thematic comparisons between Fundstat and GBARD must be interpreted in account of differences and limitations in data coverage and interpretation of directionality.

Fundstat captures R&D funding in the form of project awards, whereas GBARD reflects the total budgetary intent, including institutional funding, support delegated to funding bodies for the general advancement of knowledge (part of which can be relevant towards goals like health or climate), large R&D infrastructure investments, and support for defence-related R&D, that are not necessarily disbursed through competitive or traceable project awards (OECD, 2015^[1]).

Countries differ in the extent to which all major R&D funding agencies and programmes are included in Fundstat and the degree to which specific domains (e.g., defence, energy, or mission-oriented initiatives) are visible in project award databases may differ. As of 2021, Fundstat captures approximately 51% of the R&D funding reported in GBARD (excluding General University Funds) across the 19 OECD countries included.

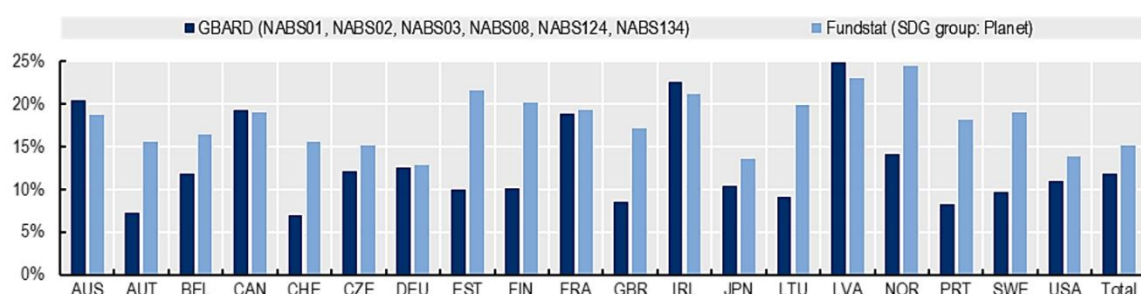
To contextualise the analysis of R&D project awards by societal goals within the OECD Fundstat database and understand potential biases relative to official statistics, a benchmark comparison is conducted against Government Budget Allocations for R&D (GBARD) for cases where Socio-Economic Objectives (SEOs) (OECD, 2025^[57]) are thematically aligned with SDGs. This is not possible for Defence which does not have a truly equivalent group in the SDG framework even though the Peace SDG is notionally aligned.

Figure A.2 provides comparisons between the share of R&D funding in the Fundstat database by country and selected SDG thematic groups, relative to the related GBARD allocation for selected SDG related clusters (Table 2.1) in 2021, the latest year for which reasonably complete GBARD data was available for this analysis:

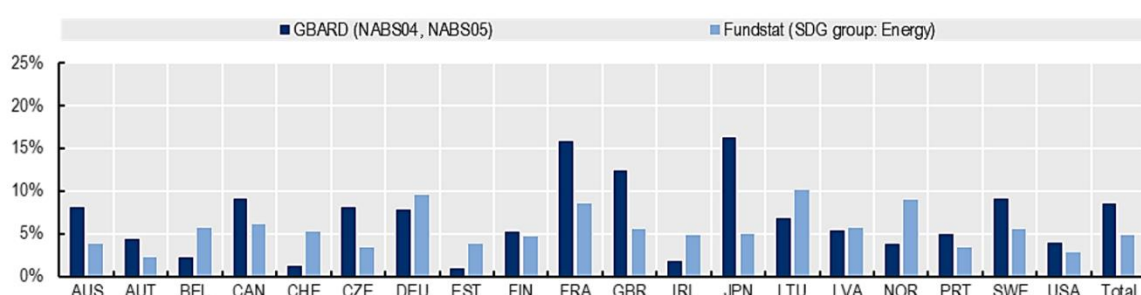
- Planet (NABS01, NABS02, NABS03, NABS08, NABS124, NABS134). This thematic area appears to be slightly over-represented in Fundstat but without a clear pattern across countries.
- Energy (NABS04, NABS05). This area appears to be somewhat under-represented in Fundstat. This may reflect that public support for Energy R&D is frequently associated with infrastructure investments, defence or security-linked programmes, and institutional support to public laboratories or utilities, which are typically not awarded as traceable project grants and therefore not captured by Fundstat.
- Health (NABS07, NABS10, NABS11, NABS123, NABS133) appears to be systematically over-represented on average by a factor of 10 points.
- Residual (NABS06, NABS121, NABS122, NABS125, NABS126, NABS131, NABS132, NABS135, NABS136, NABS14, NABS09, NABS12, NABS13). The residual includes industrial production and technology (NABS06), defence (NABS14), education (NABS09), R&D financed from general university funds (GUF) (NABS121, NABS122, NABS125, NABS126) and R&D financed from other sources than GUF (NABS131, NABS132, NABS135, NABS136). In Fundstat, the SDG groups remaining are Prosperity, Education, Peace and none, which appear to be under-represented by a factor of 10 points.

Figure A.2. Share of R&D funding by theme in Fundstat and GBARD for 19 OECD countries, 2021

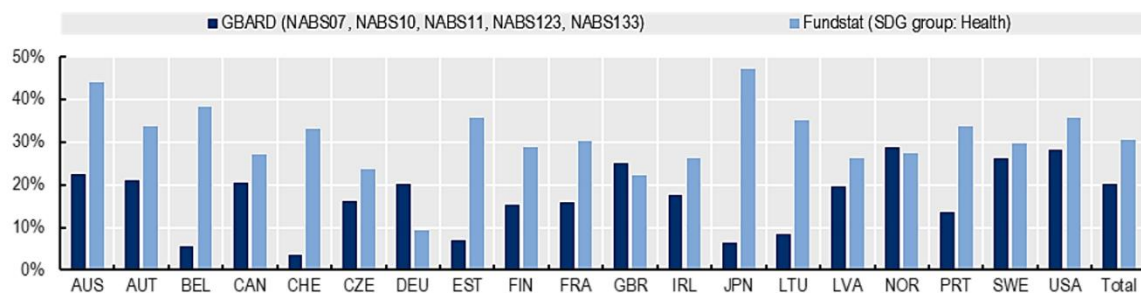
(a) Planet



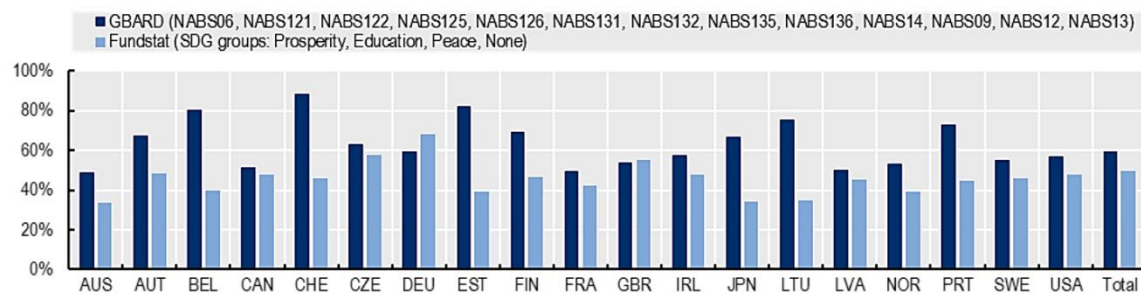
(b) Energy



(c) Health



(d) Residual



Note: Ordered by country alphabetical order. Government Budget Allocations for R&D (GBARD) provisional and estimated data exist until 2024 for some countries (OECD, 2025^[57]; OECD, 2025^[58]). The year 2021 was selected as the reference year for comparison because it represents the most recent period for which both Fundstat and GBARD offer high completeness and granularity.

Source: OECD analysis of the Fundstat database (v.2024) and the Main Science and Technology Indicators (MSTI) database (OECD, 2025^[57]; OECD, 2025^[58]).

Annex B. Supplementary methodological information

Table B.1 presents the distribution of scholarly publications labelled by Sustainable Development Goal (SDG) relevance, based on matched author-level data from the 2021 OECD International Survey of Science (ISSA3). Respondents to the ISSA survey were asked to identify the SDG most relevant to their research (see Section 3.1.1). Their responses were subsequently propagated to their scientific publications, which were identified via matched Scopus Author IDs, covering 75% of all survey respondents. The dataset comprises 11 719 publications spanning the period 2010-2021, reflecting the most recent decade of scientific output for these researchers. It includes 18 classification labels: 17 corresponding to the official SDGs and one “No SDG relevance”. The table shows the number of labelled publications split between the training dataset (10 719 samples) and a held-out validation dataset (1 000 samples), used to evaluate model generalisation during model development (see Section 4.1).

Table B.1. Distribution of labels of sustainable development goals (SDGs) across the datasets

SDG	Training dataset	Validation dataset	Total	Share of total training dataset	Share of total validation dataset	Share of total dataset (rounded to 1 d.p.)
1	40	4	44	0%	0%	0%
2	370	35	405	3%	4%	3%
3	3 006	252	3 258	28%	25%	28%
4	1 113	103	1 216	10%	10%	10%
5	48	5	53	0%	1%	0%
6	248	22	270	2%	2%	2%
7	634	44	678	6%	4%	6%
8	143	12	155	1%	1%	1%
9	1 387	148	1 535	13%	15%	13%
10	153	11	164	1%	1%	1%
11	310	34	344	3%	3%	3%
12	502	46	548	5%	5%	5%
13	802	87	889	7%	9%	8%
14	340	36	376	3%	4%	3%
15	556	55	611	5%	6%	5%
16	50	6	56	0%	1%	0%
17	105	13	118	1%	1%	1%
No SDG	912	87	999	9%	9%	9%
Total	10 719	1 000	11 719	100%	100%	100%

Note: The table shows the distribution of publications of matched 2021 ISSA respondents to their respective Scopus Author profiles across the 17 SDGs and the no SDG label (see Table 3.1).

Source: OECD analysis of scientific publications of the matched respondents of the OECD International Survey of Science (ISSA3) and the Scopus database (v.1.2024).

Model evaluation assesses how effectively a trained model performs on new, “unseen” data. This process involves applying various metrics to determine whether the model can generalise beyond the training data and produce reliable predictions. In this context, the SDG classifier, trained using labelled research outputs from the OECD ISSA3 survey, is evaluated using an external benchmark: the OSDG Community Dataset (OSDG-CD). This provides an independent test to assess the classifier’s ability to identify Sustainable Development Goal (SDG) relevance in textual excerpts not used during model development (see Section 3.2.2). The OSDG dataset consists of over 43 000 SDG-labelled excerpts, curated through crowdsourcing and expert validation. For evaluation, a high-confidence subset of 8 768 excerpts was selected, comprising only entries with full agreement (100%) among five or more volunteers and no disagreement (zero rejections).

The OECD SDG classifier was applied to this dataset to predict the most relevant SDG for each excerpt. Its performance is summarised in Figure B.1, which presents three heatmaps measuring:

- Precision (column-wise normalisation), indicating the proportion of correct predictions for each SDG among all items predicted to belong to that SDG.
- Recall (row-wise normalisation), indicating the proportion of correctly identified instances for each SDG among all true occurrences of that SDG.
- F1 score (harmonic mean of precision and recall), providing a balanced performance measure.

Overall, the classifier achieves a micro-averaged F1 score of 0.54 and a macro-averaged F1 score of 0.47, marginally outperforming existing approaches (Kashnitsky et al., 2024^[8]), underlining the added value of research relevance-based data, as collected through the ISSA2021 self-assessment survey.

As shown in Figure B.1 (panel b), the classifier demonstrates strong recall for several SDGs, including SDG 3 (Good health and well-being), SDG 4 (Quality education), SDG 9 (Industry, innovation and infrastructure), and SDG 15 (Life on land), each exceeding or approaching 90%. These results indicate the model’s strength in detecting these goals, likely due to distinctive language patterns and abundant, high-quality training data. SDGs 2, 6, 7, and 16 also show solid recall performance. In contrast, some SDGs (e.g., SDG 1, 5, 8, and 12) exhibit lower recall, possibly due to overlapping thematic content or less clearly defined text. Precision patterns Figure B.1 (panel a), differ slightly, with SDGs 6, 9, 10, 13, 14, 15, and 16 showing particularly high scores. For example, SDG 6 achieves a precision of 90%, meaning the classifier is highly accurate when it predicts this SDG. These results suggest that when the classifier assigns certain SDGs, it does so with high confidence and low false-positive rates. The F1 score, shown in Figure B.1 (panel c), provides a holistic view of classifier performance. The strongest alignment is observed along the diagonal, indicating correct label matches. SDGs with high F1 scores reflect both good recognition (recall) and precision, while lower F1 scores often reveal an imbalance, either the classifier frequently misses the correct label or mislabels content as belonging to that SDG. SDG 0 (No SDG relevance) also appears as a classifier prediction within the OSDG dataset. From an evaluation perspective, the classifier very rarely assigns this label, reflecting a low tendency to predict irrelevance. This behaviour is consistent with the high-quality nature of the benchmark dataset, which was constructed to include only excerpts with strong agreement among raters.

Performance differences across SDGs can be attributed to a mix of factors, including the clarity and distinctiveness of goal-related terminology, training data volume, and the presence of interdisciplinary or ambiguous content.

Figure B.1. Out-of-sample SDG label classifier prediction on the OSDG dataset (n=8 768)

(a) Precision (column-wise normalisation)

SDG label based on classifier prediction

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	0
1	0.0	0.2	0.2	0.0	0.2	0.0	0.0	0.1	0.2	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	0.0	0.7	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3	0.0	0.0	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	0.0	0.0	0.0	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5	0.1	0.1	0.4	0.1	0.2	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6	0.0	0.1	0.1	0.0	0.0	0.6	0.0	0.0	0.0	0.0	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0
7	0.0	0.0	0.0	0.0	0.0	0.0	0.6	0.0	0.2	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0
8	0.2	0.0	0.1	0.1	0.1	0.0	0.0	0.2	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
10	0.1	0.1	0.1	0.0	0.2	0.0	0.0	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
11	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
12	0.0	0.0	0.1	0.0	0.0	0.0	0.1	0.0	0.4	0.0	0.0	0.3	0.0	0.0	0.0	0.0	0.0	0.0
13	0.0	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.2	0.0	0.1	0.0	0.4	0.0	0.0	0.0	0.0	0.0
14	0.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.0	0.0	0.0	0.0
15	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.8	0.0	0.0	0.0
16	0.0	0.0	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.0	0.1

(b) Recall (row-wise normalisation)

SDG label based on classifier prediction

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	0
1	0.0	0.2	0.0	0.0	0.1	0.0	0.0	0.1	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0
3	0.0	0.0	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	0.0	0.0	0.0	0.8	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5	0.3	0.1	0.3	0.1	0.4	0.0	0.0	0.2	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.0
6	0.0	0.0	0.0	0.0	0.0	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7	0.0	0.0	0.0	0.0	0.0	0.0	0.8	0.0	0.1	0.0	0.1	0.2	0.3	0.0	0.0	0.0	0.0	0.0
8	0.2	0.0	0.0	0.0	0.1	0.0	0.0	0.1	0.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
10	0.2	0.0	0.0	0.0	0.1	0.0	0.0	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
11	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0
12	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.4	0.0	0.0	0.0	0.0	0.0	0.0
13	0.0	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.1	0.0	0.1	0.1	0.5	0.0	0.0	0.0	0.0	0.0
14	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.8	0.0	0.0	0.0	0.0
15	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.8	0.0	0.0	0.0	0.0
16	0.1	0.0	0.1	0.0	0.3	0.0	0.0	0.1	0.0	0.5	0.1	0.0	0.0	0.0	0.1	0.9	0.0	0.9

(c) F1 score (harmonic mean between precision and recall)

SDG label based on classifier prediction

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	0
1	0.0	0.2	0.1	0.0	0.1	0.0	0.0	0.1	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	0.0	0.3	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0
3	0.0	0.0	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	0.0	0.0	0.0	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5	0.1	0.1	0.3	0.1	0.3	0.0	0.0	0.1	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6	0.0	0.0	0.0	0.0	0.0	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7	0.0	0.0	0.0	0.0	0.0	0.0	0.7	0.0	0.1	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0
8	0.2	0.0	0.0	0.0	0.1	0.0	0.0	0.2	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
10	0.2	0.0	0.1	0.0	0.2	0.0	0.0	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
11	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0
12	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.4	0.0	0.0	0.0	0.0	0.0	0.0
13	0.0	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.5	0.0	0.0	0.0	0.0	0.0
14	0.0	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.5	0.0	0.0	0.0	0.0
15	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.8	0.0	0.0	0.0
16	0.0	0.0	0.1	0.0	0.2	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.6	0.0	0.2

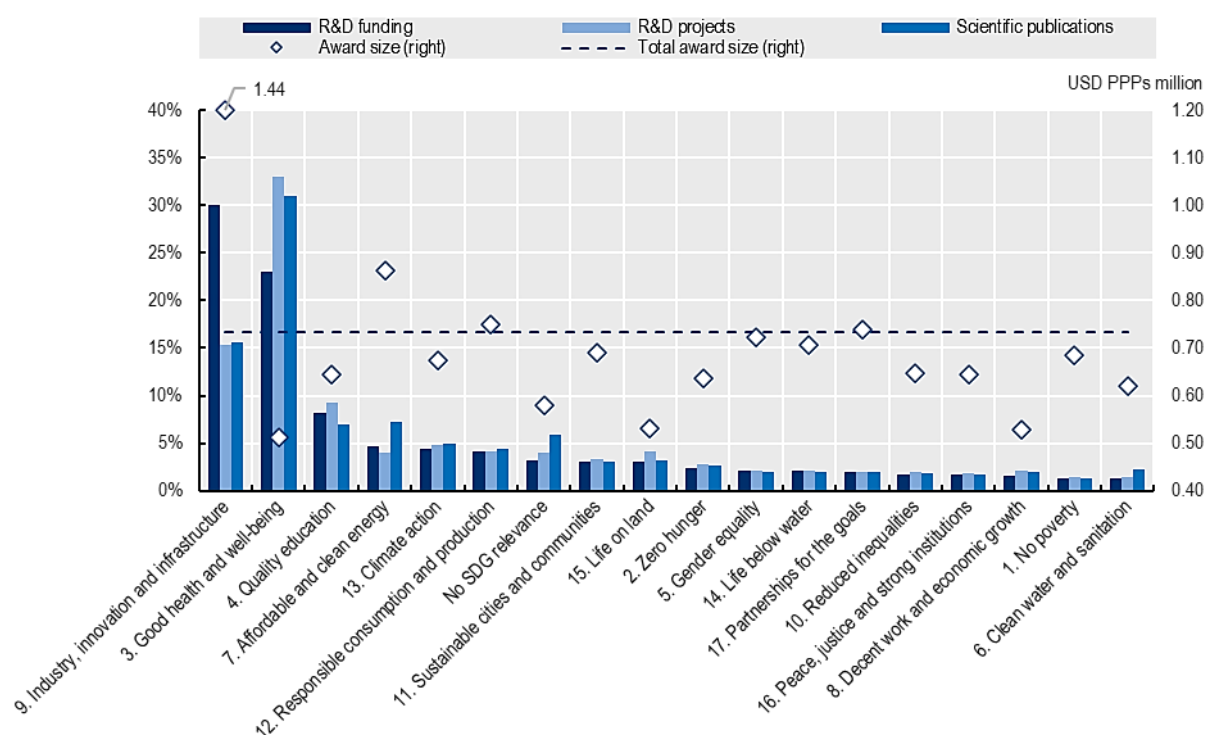
Note: The OSDG Community Dataset (OSDG-CD) is a public dataset that contains SDG-labelled text excerpts, coming from various sources, validated by citizen scientists with respect to SDGs. The original dataset has 43 000 text excerpts and does not contain excerpts with labels SDG 17 or No SDG relevance (SDG 0). The OECD STI team restricted the sample to 8 768 text excerpts representing text with: (i) perfect agreement among volunteers (100%), (ii) zero negative labels (rejections), and (iii) which 5 or more volunteers have agreed with.

Source: OECD analysis of the OECD SDG classifier with the OSDG dataset (OSDG; UNDP IICPSD SDG AI Lab; PPMI, 2024^[56]).

Annex C. Supplementary results

Figure C.1. SDG distribution of R&D projects, R&D funding and scientific publications, 2015-2023

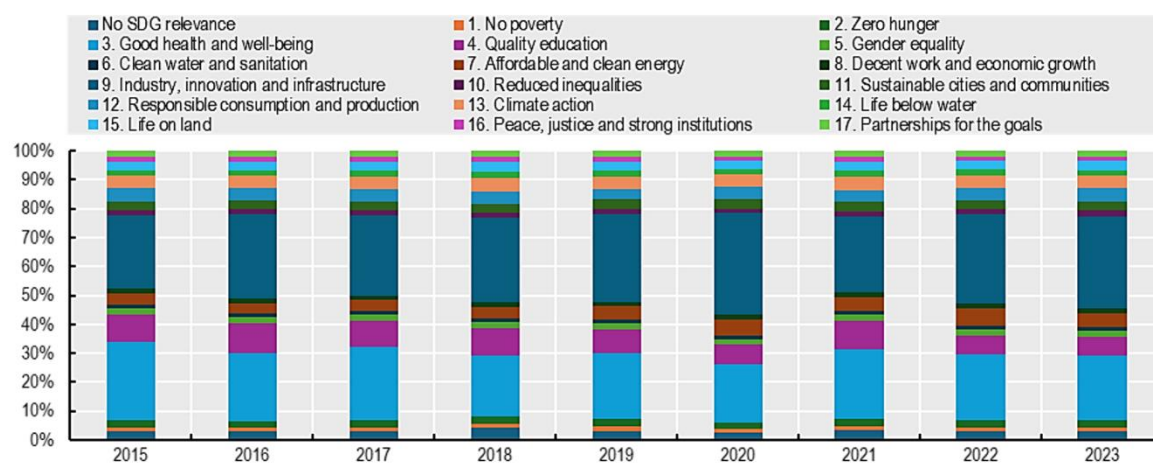
As a share of total R&D funding in the OECD Fundstat database (19 OECD countries and EU-EC programmes)



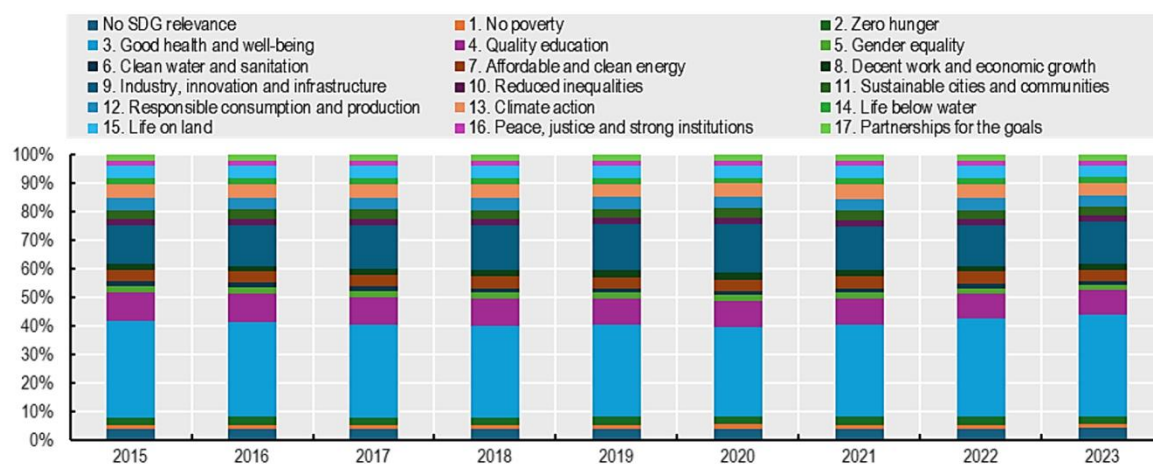
Source: OECD analysis of the Fundstat database (v.2024) and Scopus database (v.1.2024).

Figure C.2. Distribution of R&D funding and R&D projects by SDG, 2015-2023

(a) Share of R&D funding in total R&D funding, by SDG



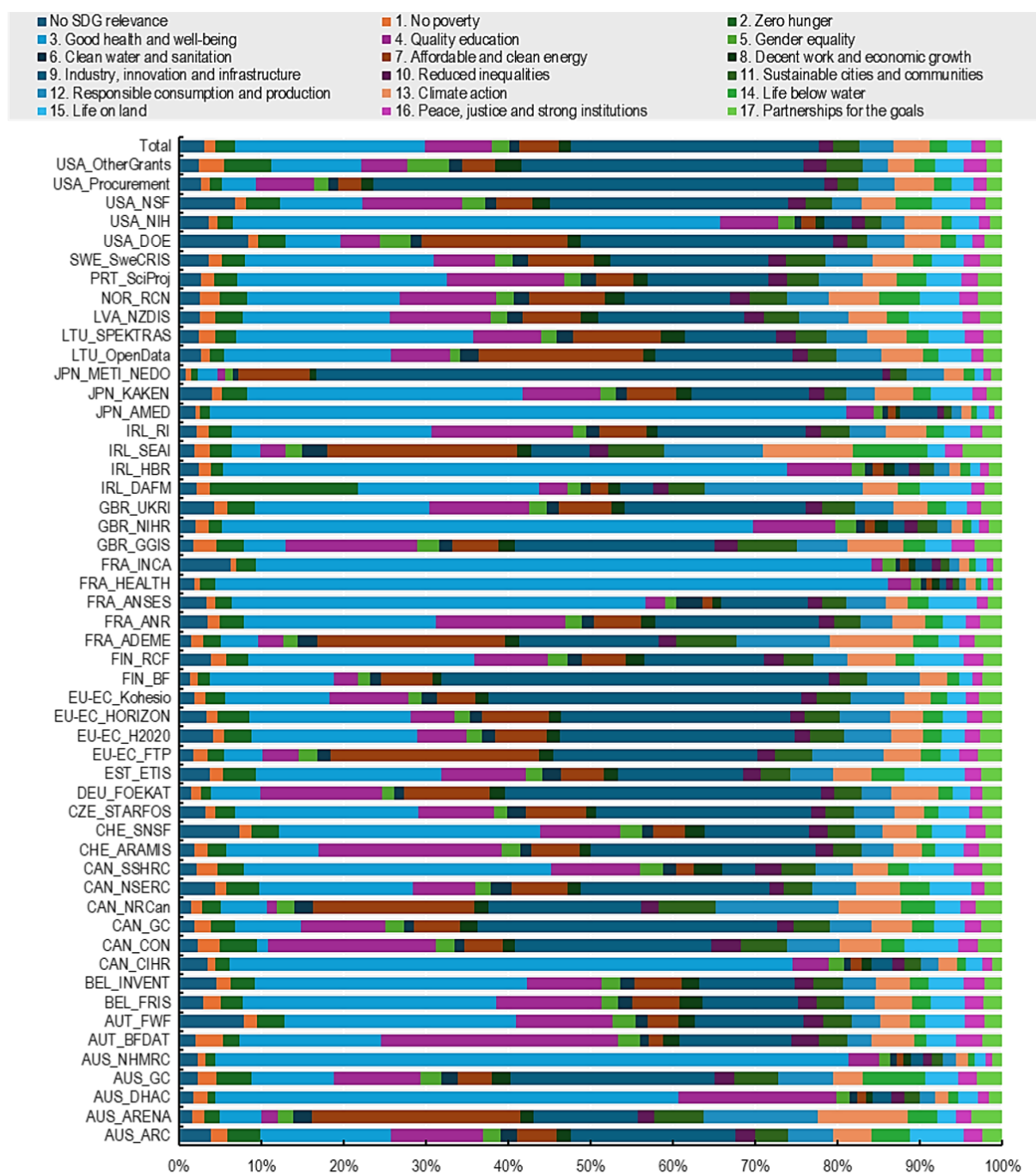
(b) Share of R&D projects in total R&D projects, by SDG



Note: The analysis excludes projects with missing or nil funding information, which obey disclosure controls or database entries intended to register project modifications without funding implications (zero additional funding). Refer to Table 5.1 for the definition of SDG groups.

Source: OECD analysis of Fundstat database (v.2024).

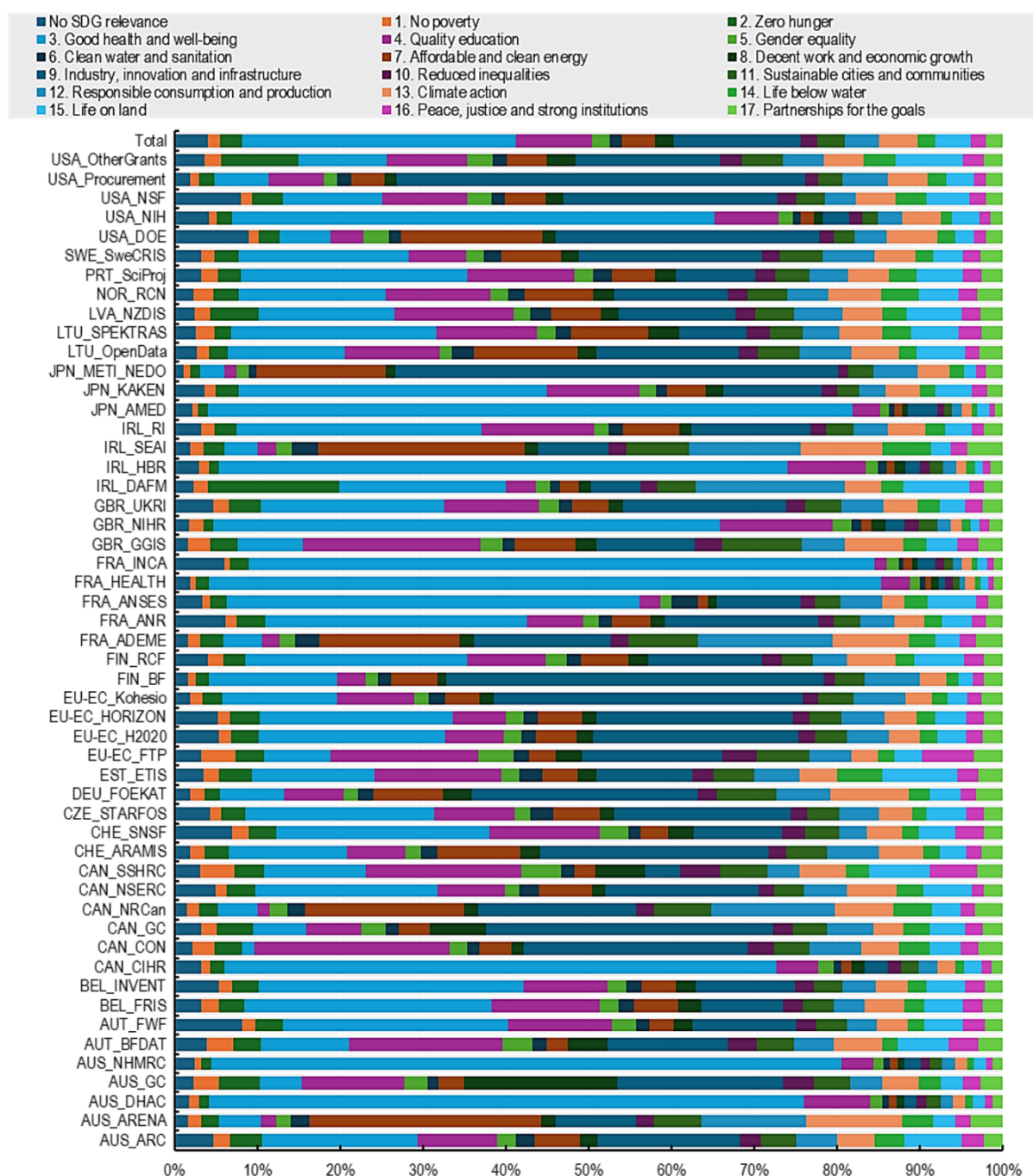
Figure C.3. Distribution of R&D funding by SDG & data source, 2015-2023



Note: Ordered by country/area code followed by data source in alphabetical order in descending order. The analysis excludes projects with missing or nil funding information, which obey disclosure controls or database entries intended to register project modifications without funding implications (zero additional funding). See Annex B for data source names, and Section 5.1.1 for data source comparisons on R&D funding.

Source: OECD analysis of Fundstat database (v.2024).

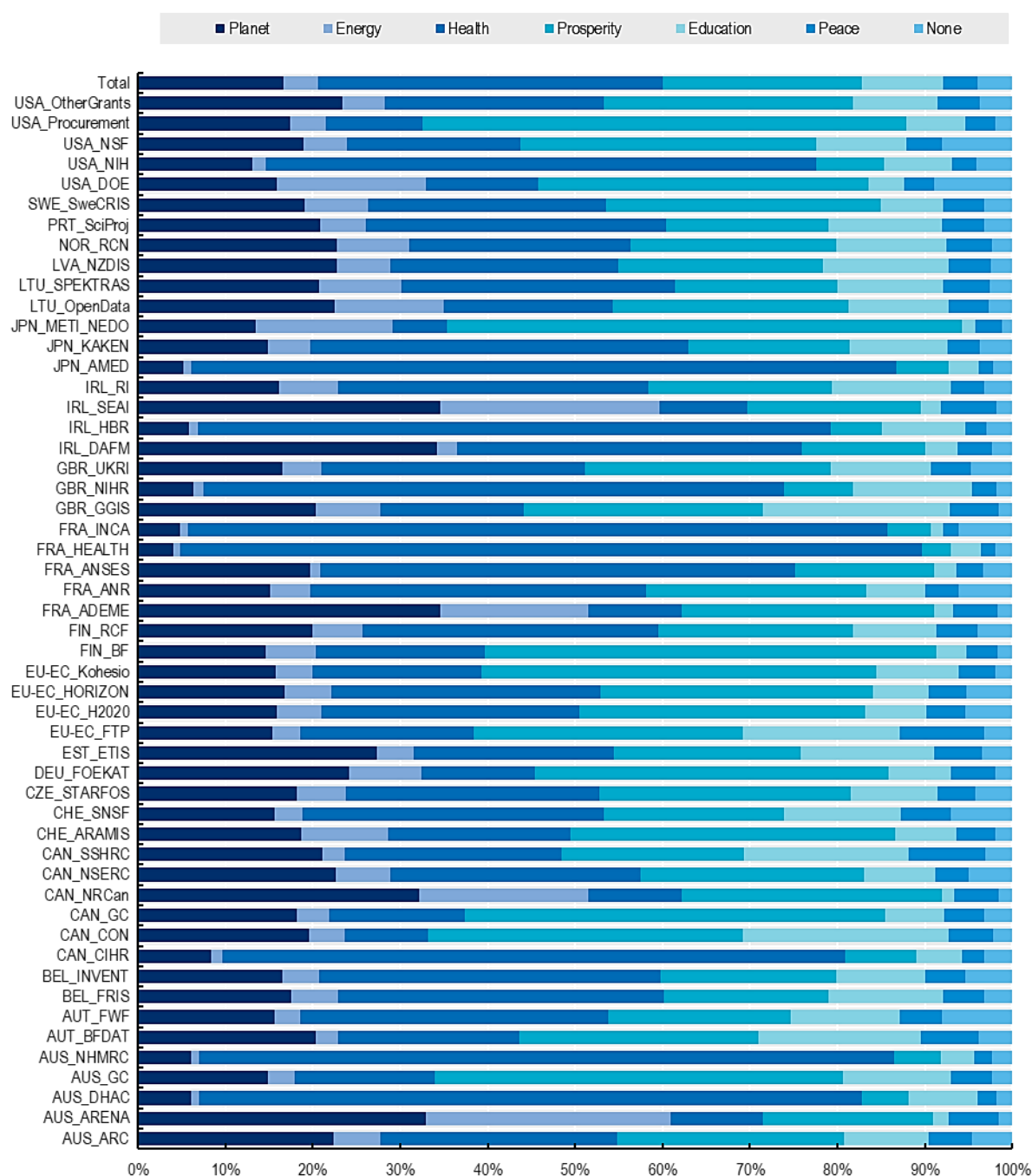
Figure C.4. Distribution of R&D projects by SDG & data source, 2015-2023



Note: Ordered by country/area code followed by data source in alphabetical order in descending order. The analysis excludes projects with missing or nil funding information, which obey disclosure controls or database entries intended to register project modifications without funding implications (zero additional funding). See Annex B for data source names, and Section 5.1.1 for data source comparisons on R&D funding.

Source: OECD analysis of Fundstat database (v.2024).

Figure C.5. SDG group distribution of R&D projects by data source, 2015-2023

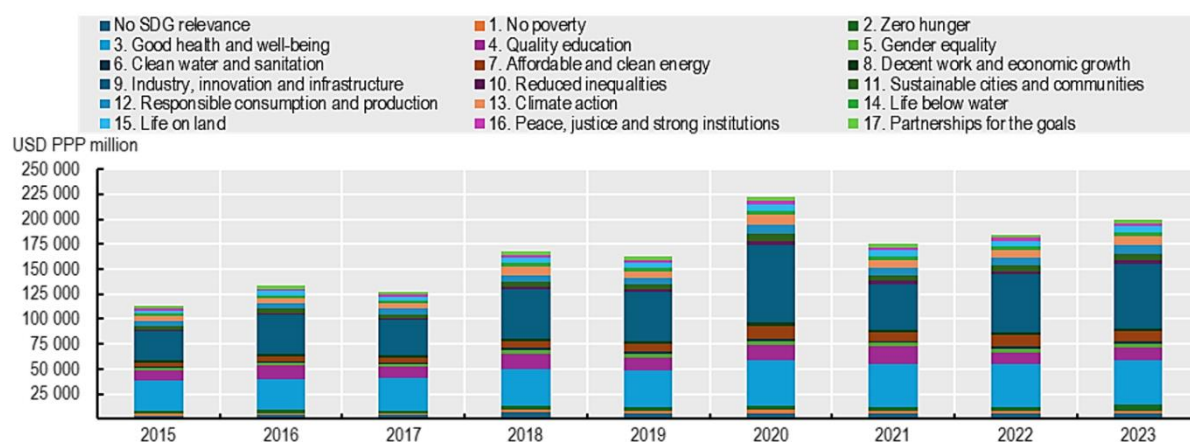


Note: Ordered by country/area code alphabetical order in descending order. The analysis excludes projects with missing or nil funding information. See Annex A for data source names, and Section 5.1.1 for data source comparisons on R&D funding.

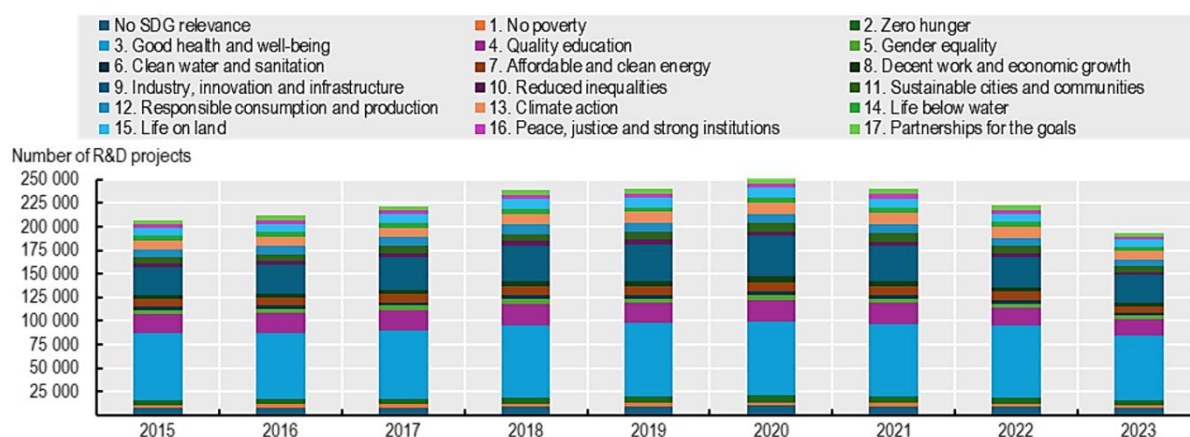
Source: OECD analysis of Fundstat database (v.2024).

Figure C.6. Total R&D funding, R&D projects and scientific publications by SDG, 2015-2023

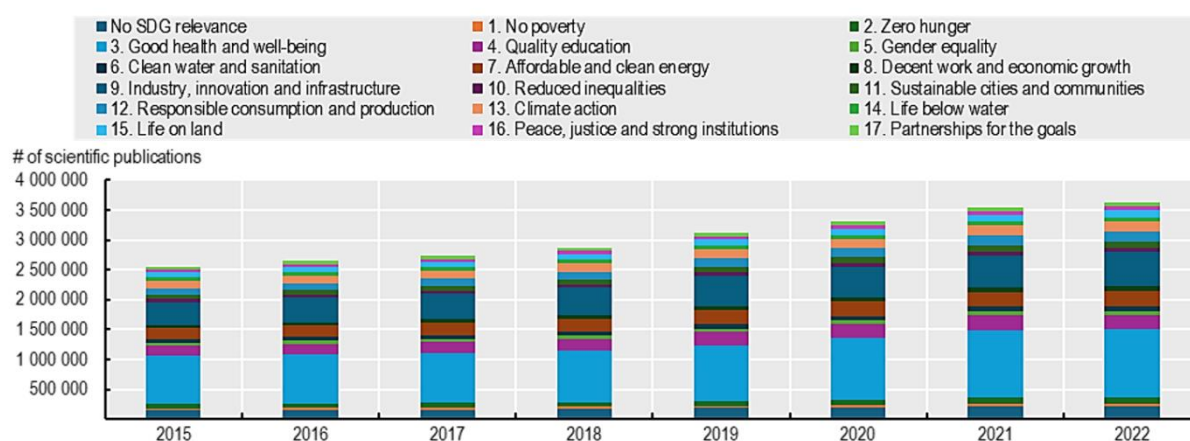
(a) Total R&D funding (USD PPP million)



(b) Total number of R&D projects



(c) Total scientific publications by SDG (2015-2022)



Note: The analysis excludes projects with missing or nil funding information, which obey disclosure controls or database entries intended to register project modifications without funding implications (zero additional funding). The Elsevier's Scopus database is constrained to publication articles, review articles, and conference papers.

Source: OECD analysis of the Fundstat database (v.2024) and Scopus database (v.1.2024).